



IT3021 – Data Warehousing and Business Intelligence
BSc (Hons) in Information Technology Specialized in Data
Science

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Assignment 01
Report

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Dataset: [Supply Chain and Sales Datasets](#)

Contents

Task 1: Dataset Selection	3
Overview	3
Dataset Information.....	4
Task 2: Preparation of data sources.....	5
Task 3: Solution architecture	9
Solution Diagram DW & BI.....	9
Architectural Components.....	9
Architecture Diagram.....	11
Task 4: Data warehouse design & development	12
Schema Description	12
Assumptions	12
Star Schema Relational Diagram	13
Task 5: ETL Development.....	17
Order of Execution and Task Details	17
1. Data Extraction	17
2. Data Transformation	17
3. Data Loading	18
4. Error Handling.....	19
Execution Order Summery	19
Task 6: ETL development – Accumulating fact tables	29
Process Overview	29

Task 1: Dataset Selection

Overview

The dataset I used in this project is a retail supply chain and sales transactional dataset, which can be used in data warehousing and analytics. It captures detailed information about customer orders, products, sales, shipping, and delivery, reflecting real-world retail operations and supporting both operational and analytical needs in business intelligence.

Scenario: This dataset represents a retail company that manages customer orders, product sales, and delivery operations through a supply chain system. Customers place orders for different products, which are then processed, shipped, and delivered using various shipping methods. The company tracks sales performance, delivery timelines, and profitability to improve operational efficiency and customer satisfaction.

Scope: It tracks customer orders, products, sales performance, discounts, profits, shipping modes, and delivery timelines over one year, enabling analysis of sales trends, customer behaviour, product demand, and supply chain efficiency.

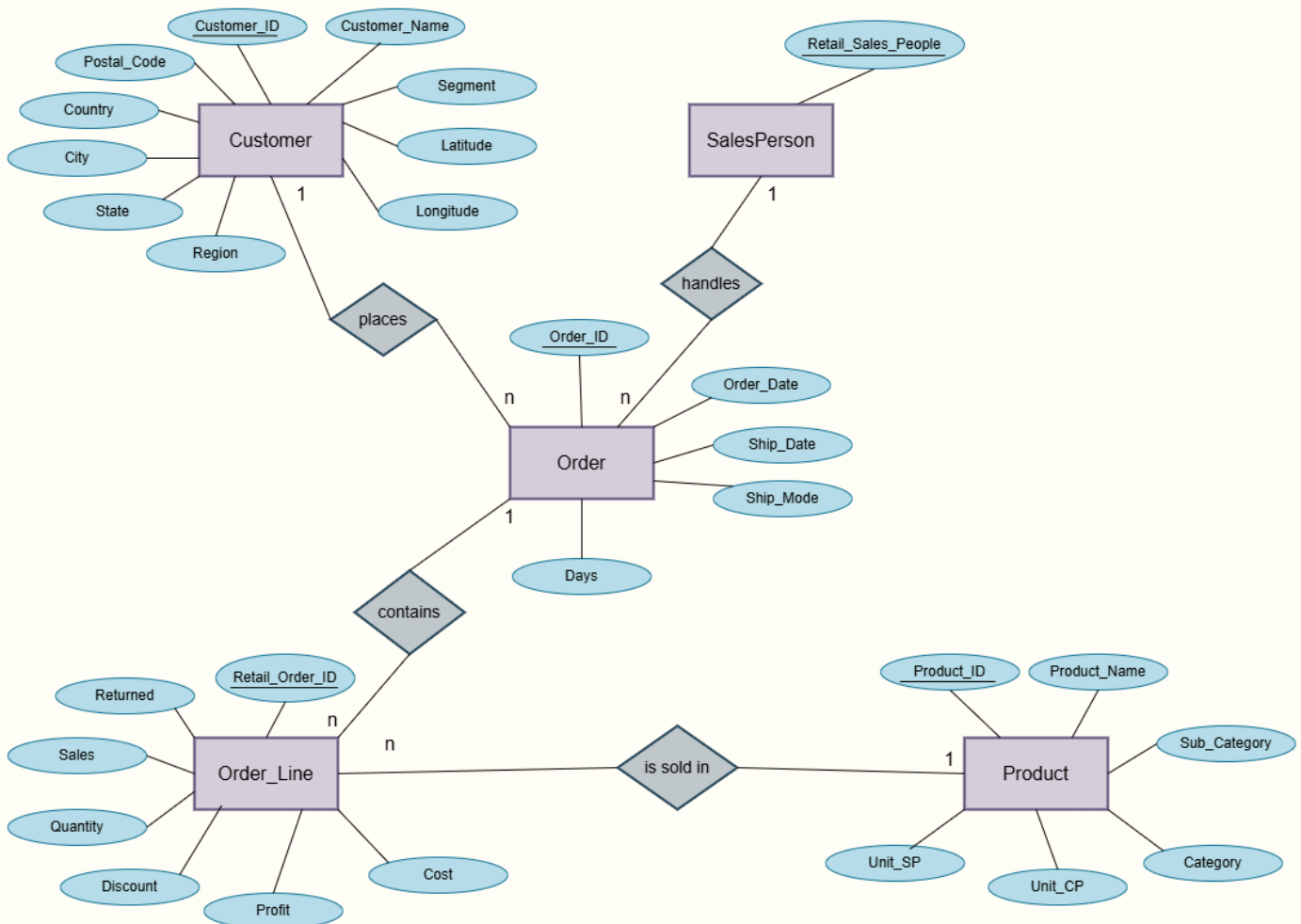


Figure 1 ER Diagram

Justification: This dataset is suitable for the assignment as it meets all the requirements of an OLTP dataset with sufficient attributes and relationships. It contains transactional data over a period, enabling time-based analysis. The dataset supports multiple entities such as customers, products, and orders, which can be transformed into dimensions and fact tables for data warehouse design. Additionally, it can be easily divided into multiple data sources (e.g., CSV, Excel, database) to demonstrate ETL processes. The presence of hierarchical attributes (such as product category and sub-category) and measurable values (sales, profit, quantity) makes it ideal for building a complete data warehousing and business intelligence solution.

Dataset Information

Data Volume

- **Time Coverage:** Retail sales orders span 3 January 2014 – 30 December 2017 (exactly 4 full year).
- **Records:**
 - 793 unique customers
 - 1,850 unique products
 - 5,009 unique orders
 - 9,994 order_Lines (6–7 order lines per day on average)
 - 4 Retail_SalesPeople
 - 3 main categories & 17 sub-categories
 - 49 US states + District of Columbia (geographic coverage)

Data Warehouse Design

- **Star Schema**
 - **Fact Table:** Fact_OrderLine (Retail_Order_ID, Order_ID, Date_SK, Customer_SK, Product_SK, SalesPerson_SK, Quantity, Sales, Discount, DiscountAmount, Cost, Profit, ReturnedFlag, DaysToShip)
 - **Dimension Tables:** Dim_Customer, Dim_Product, Dim_SalesPerson, Dim_Date

Task 2: Preparation of data sources

To support a realistic ETL process and effectively demonstrate the integration of multiple heterogeneous data sources using SSIS, the original single Excel dataset was logically divided into three distinct source types. This separation was performed based on the entities and relationships identified in the ER diagram, ensuring proper data organization, improved data handling, and alignment with real-world data warehousing practices.

- ***customer_master (Flat File / CSV):***

Description: This source contains master data of the Customer entity extracted from the ER diagram.

Attributes: Customer_ID, Customer_Name, Segment, Postal_Code, Country, City, State, Region, latitude, longitude (10 attributes).

- ***product_master (Excel File):***

Description: This source contains master data of the Product entity extracted from the ER diagram.

Attributes: Product_ID, Product_Name, Sub_Category, Category, Unit_CP, Unit_SP (6 attributes).

- ***order_transactions.db (SQL Database):***

Description: This source contains the core transactional data linked to Order, Order_Line, and SalesPerson entities and their relationships (“places”, “contains”, “handles”, “is sold in”).

Attributes: Retail_Order_ID, Order_ID, Order_Date, Ship_Date, Ship_Mode, Days, Customer_ID, Retail_Sales_People, Product_ID, Returned, Sales, Quantity, Discount, Profit, Cost (15 attributes).

Overall Justification

These three data sources were intentionally derived from the original dataset while strictly adhering to the structure defined in the ER diagram. The use of multiple source types (CSV, Excel, and Database) not only exceeds the minimum requirement of two sources but also enables the demonstration of various SSIS connection managers, including Flat File, Excel, and OLE DB. Furthermore, this approach preserves the relationships identified in the ER diagram and facilitates the preparation of clean, structured master data for dimension tables, while maintaining raw transactional data for accurate fact table population. **Transactional data was exported from SQLite to CSV and imported into SQL Server database Retail_OLTP_Source using SSMS Import Flat File wizard. This provides an OLE DB database source for SSIS ETL processes.**

Name	Date modified	Type	Size
customer_master	21/03/2026 14:22	Microsoft Excel Comma Separated Values File	79 KB
order_transactions	21/03/2026 14:22	Data Base File	1,588 KB
prepare_sources.py	21/03/2026 14:20	Python File	2 KB
product_master	21/03/2026 14:22	Microsoft Excel Worksheet	112 KB
Supply_Chain_and_Sales_Datasets	05/03/2026 21:18	Microsoft Excel Worksheet	1,857 KB

Figure 2 Data Sources

```

import sqlite3

import pandas as pd

# Read original Excel
df = pd.read_excel("Supply_Chain_and_Sales_Datasets.xlsx", sheet_name="Retail Order")

# Extract Customer attributes (based on ER diagram)
customer_cols = ['Customer ID', 'Customer Name', 'Segment', 'Postal Code',
                 'Country', 'City', 'State', 'Region', 'latitude', 'longitude']
df_customer = df[customer_cols].drop_duplicates(subset=['Customer ID'])

# Save as CSV
df_customer.to_csv("customer_master.csv", index=False, encoding='utf-8')
print(f"Customer source created: {len(df_customer)} unique customers")

# Extract Product attributes (based on ER diagram)
product_cols = ['Product ID', 'Product Name', 'Sub-Category', 'Category',
               'Unit CP', 'Unit SP']
df_product = df[product_cols].drop_duplicates(subset=['Product ID'])

# Save as Excel file (different source type)
df_product.to_excel("product_master.xlsx", index=False, sheet_name="Product")
print(f"Product source created: {len(df_product)} unique products")

# Keep all transactional columns + relationships
order_cols = ['Retail Order ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
             'Days', 'Customer ID', 'Retail Sales People', 'Product ID',
             'Returned', 'Sales', 'Quantity', 'Discount', 'Profit', 'Cost']
df_orders = df[order_cols]

# Create SQLite database and table
conn = sqlite3.connect("order_transactions.db")
df_orders.to_sql("order_lines", conn, index=False, if_exists="replace")
conn.close()

print("Database source created: order_transactions.db with table 'order_lines'")
print(f"Records loaded: {len(df_orders)}")

```

Figure 3 python code used to separate data Sources

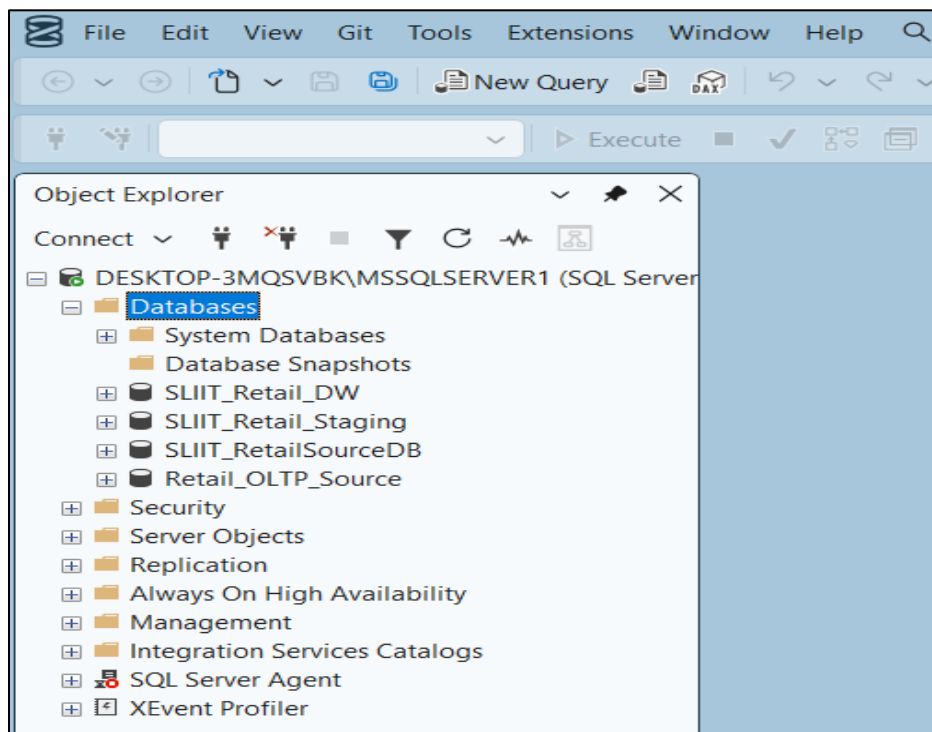


Figure 4 Create Source Database

Import Source_OrderLine sources file to Retail_OLTP_Source Database

Import Flat File 'Retail_OLTP_Source'

Modify Columns

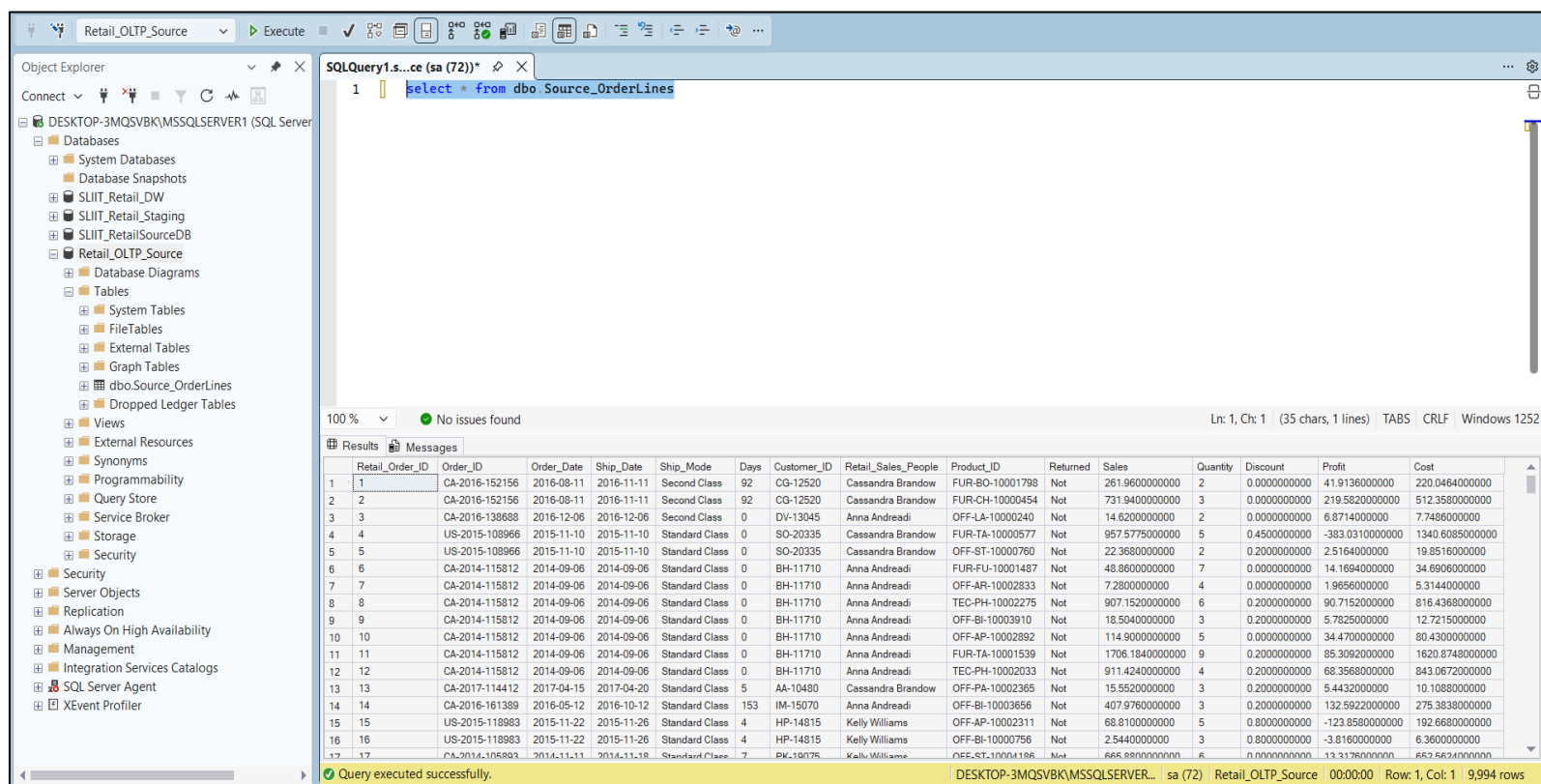
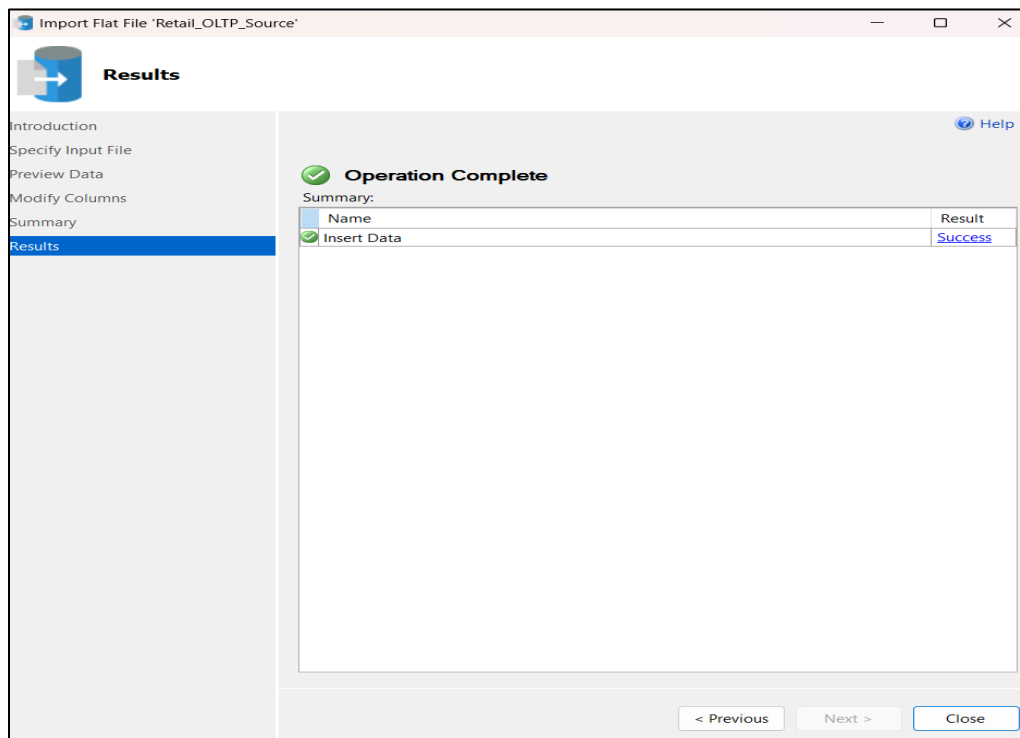
[Introduction](#) [Specify Input File](#) [Preview Data](#) [Modify Columns](#) [Summary](#) [Results](#) [Help](#)

Modify Columns
This operation generated the following table schema. Please verify if schema is accurate, and if not, please make any changes.

Column Name	Data Type	Primary Key	Allow Nulls
Retail_Order_ID	int	☐	☐
Order_ID	nvarchar(50)	☐	☐
Order_Date	date	☐	☐
Ship_Date	date	☐	☐
Ship_Mode	nvarchar(50)	☐	☒
Days	int	☐	☒
Customer_ID	nvarchar(50)	☐	☐
Retail_Sales_People	nvarchar(50)	☐	☒
Product_ID	nvarchar(50)	☐	☐
Returned	nvarchar(50)	☐	☒
Sales	decimal(18, 10)	☐	☐
Quantity	int	☐	☐
Discount	decimal(18, 10)	☐	☐
Profit	decimal(18, 10)	☐	☐
Cost	decimal(18, 10)	☐	☐

Row granularity of error reporting (performance impact with smaller ranges) No Range

< Previous Next > Cancel

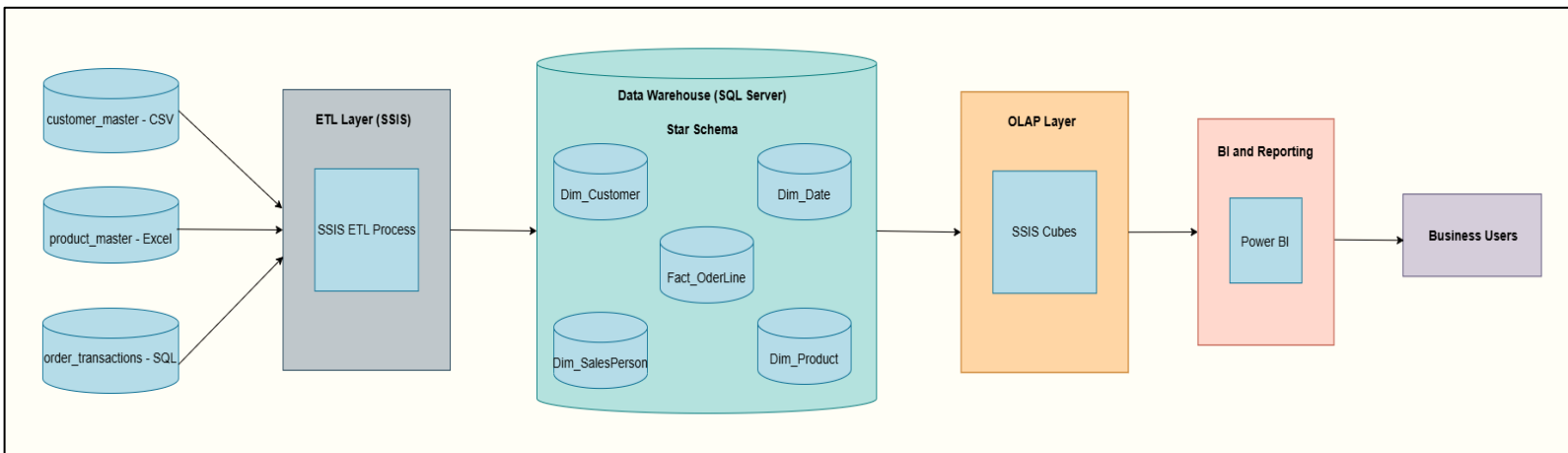


- The source database Retail_OLTP_Source contains only one table (Source_OrderLines) with raw transactional data. Customer master data is kept as CSV and Product master data as Excel to maintain three heterogeneous source types as required by the assignment.

Task 3: Solution architecture

The proposed solution follows a classic Microsoft DW/BI layered architecture that transforms the retail OLTP dataset (Supply_Chain_and_Sales_Datasets.xlsx) into a fully functional Data Warehouse and Business Intelligence system. The architecture consists of heterogeneous data sources, a staging area, ETL processes using SSIS, a star-schema Data Warehouse in SQL Server, OLAP cubes in SSAS, and BI tool, power BI is used for reporting and visualization.

Solution Diagram DW & BI



Architectural Components

1. Data Sources

- CSV (customer_master – Customer entity)
- Excel (product_master – Product entity)
- SQL Database (order_transactions – Order, Order_Line & SalesPerson entities).

2. ETL Layer

- **Tool:** SQL Server Integration Services (SSIS)
- **Tasks:** Extract, transform, and load data from the three different sources into the Data Warehouse (includes Flat File Source, Excel Source, OLE DB Source, SCD Type 2 on Customer, lookups, and calculations).

3. Data Warehouse

- **Platform:** SQL Server
- **Schema:** Star schema for its simplicity and query speed, which is suitable for our retail sales reporting needs (Fact_OrderLine + Dim_Date, Dim_Customer (SCD Type 2), Dim_Product, Dim_SalesPerson).

4. OLAP / Analytics Layer

- **Tool:** SQL Server Analysis Services (SSAS)
- **Purpose:** Cubes for multidimensional analysis (hierarchies, aggregates, KPIs, and measures such as Sales, Profit, Quantity and Return Rate).

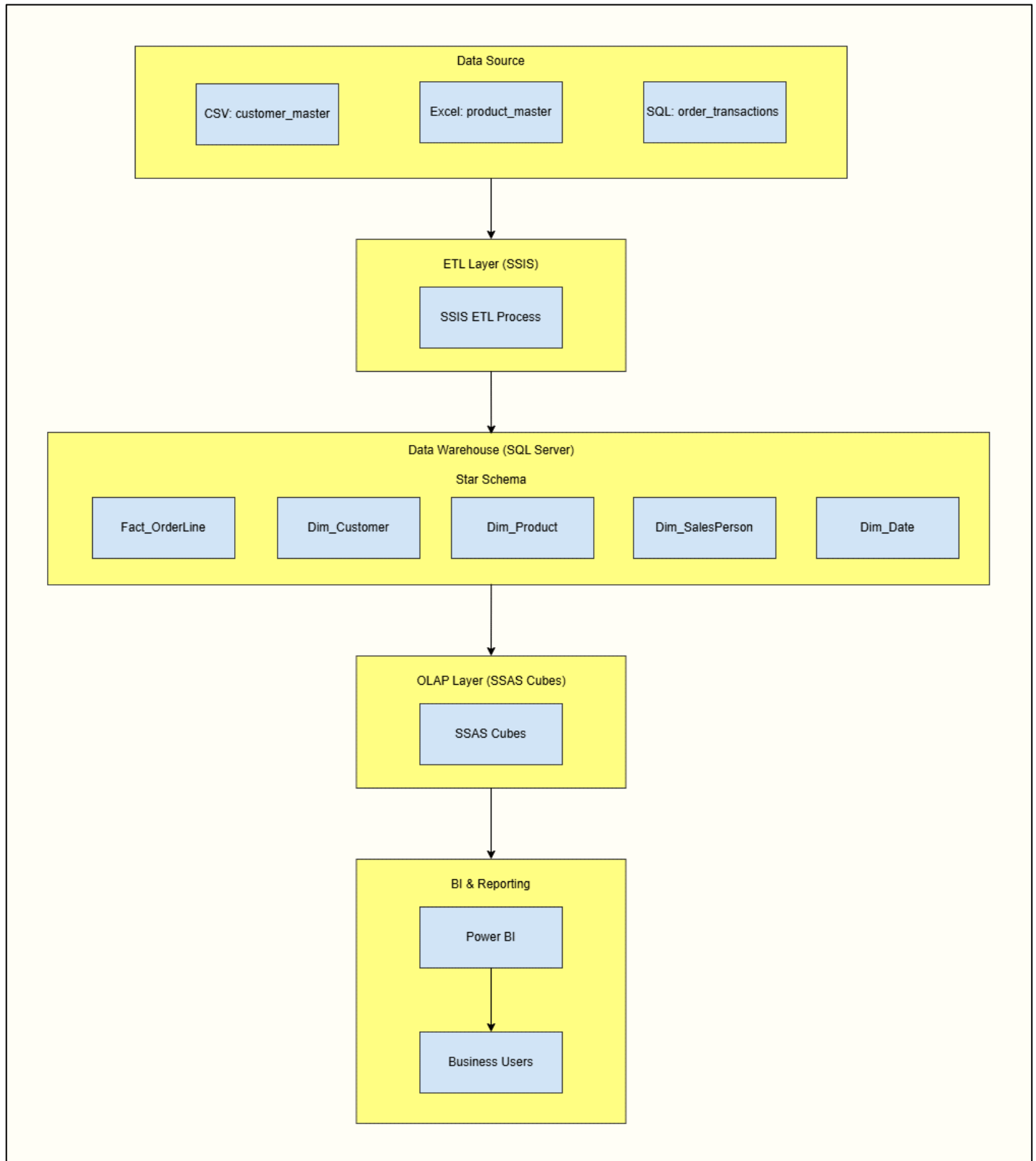
5. BI and Reporting

- **Tool:** Power BI
- **Purpose:** Dashboards, ad-hoc reports, and analytics (sales by region/category/salesperson, profit trends, return rate analysis, shipping performance, etc.).

Architecture Diagram Components

Component	Description
Data Sources	Three heterogeneous sources created in Task 2: customer_master.csv (Customer entity), product_master.xlsx (Product entity), and order_transactions.db (Order, Order_Line & SalesPerson entities).
ETL (SSIS)	SQL Server Integration Services packages that extract data from the three different source types, perform transformations (SCD Type 2, lookups, calculations), and load data into the star schema.
Data Warehouse	SQL Server database containing a star schema (Fact_OrderLine + 4 dimension tables including Dim_Customer as SCD Type 2) for analytics and reporting.
OLAP (SSAS)	SQL Server Analysis Services multidimensional cube built on the star schema with hierarchies (Date, Product, Geography), measures (Sales, Profit, Quantity), and KPIs.
BI Tools	Enables users to create dashboards, reports, and perform ad-hoc analysis.

Architecture Diagram



Task 4: Data warehouse design & development

Dimensional Model: Star Schema

A star schema dimensional model has been designed and implemented in SQL Server for the Retail Sales dataset. Star schema was chosen over snowflake for its simplicity, faster query performance, and suitability for SSAS cube development and reporting.

Fact Table (one):

- Fact_OrderLine

Dimension Table (four)

- Dim_Date
- Dim_Customer (SCD Type 2)
- Dim_Product
- Dim_SalesPerson

Schema Description

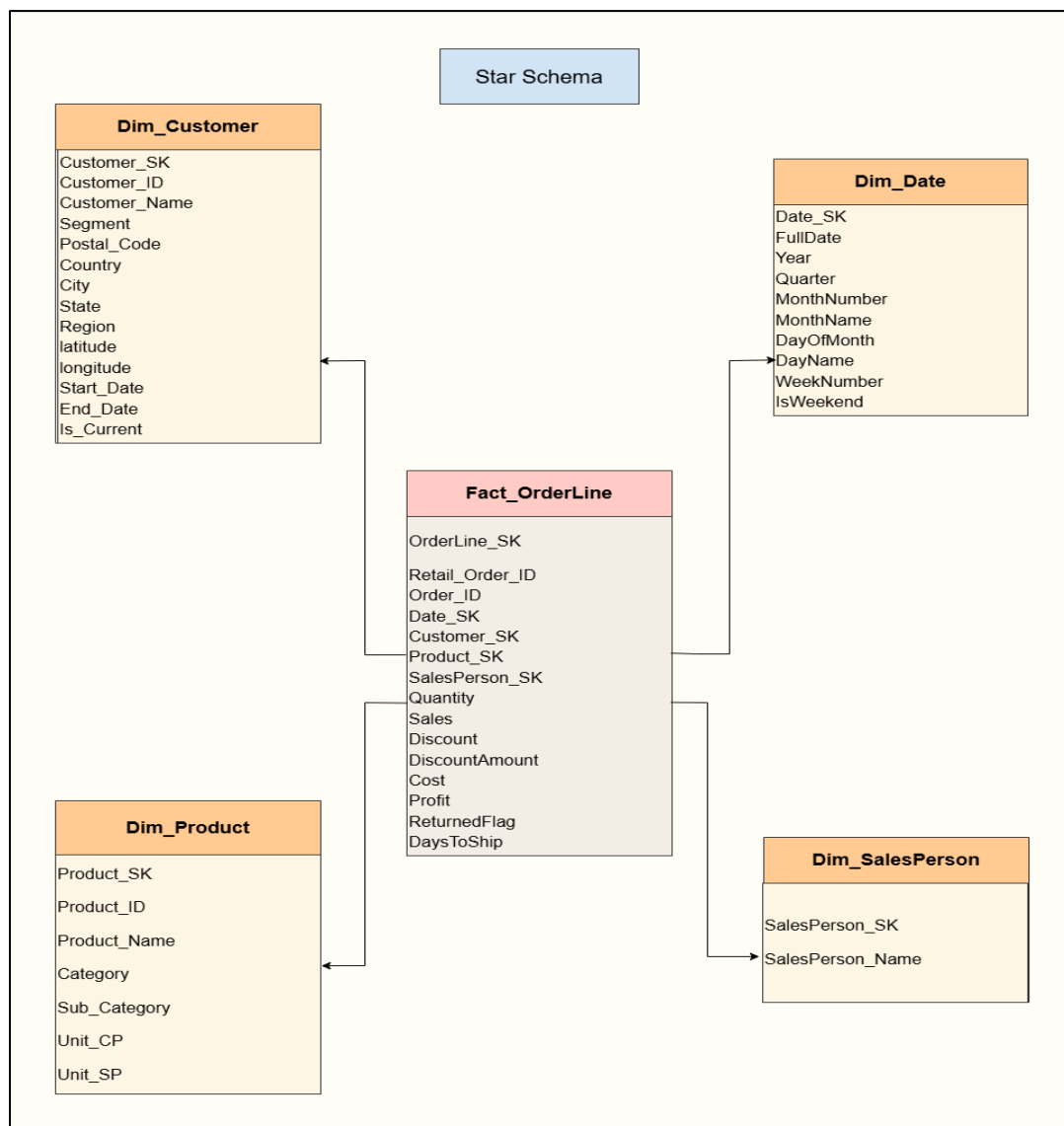
- **Star Schema:** Centralized Fact_OrderLine linked to dimensions.
- **Dimensions:**
 - Dim_Date: Date hierarchy for temporal analysis (Year → Quarter → Month → Day) to support time-based trends, year-over-year comparisons, and shipping delay calculations.
 - Dim_Customer: Tracks customer information with Type 2 SCD for historical changes in address, region, segment, and location (latitude/longitude) over time.
 - Dim_Product: Product master data with hierarchy (Category → Sub_Category) for product-level analysis; uses Type 1 SCD (static attributes like name, cost price, and selling price rarely change).
 - Dim_SalesPerson: Salesperson details (static, Type 1 SCD) to track performance, regional assignments, and commission-related reporting.

Assumptions

1. Fact table (Fact_OrderLine) grain is at order-line level (one row per line item), exactly matching the raw OLTP source data.

2. Dim_Customer uses SCD Type 2 to track historical changes in address, region, segment, and location.
3. Dim_Product and Dim_SalesPerson use SCD Type 1 (static attributes that rarely change).
4. Dim_Date covers 2013–2018 to support complete time hierarchies (Year → Quarter → Month → Day).
5. Product hierarchy (Category → Sub_Category) and Customer geographic hierarchy (Country → Region → State → City) are included for drill-down analysis.
6. Star schema (instead of snowflake) is chosen for simplicity and fast query performance in SSAS cubes.
7. Monetary measures use DECIMAL(18,4) for precision; Returned status is converted to BIT flag.
8. Degenerate dimensions (Order_ID, Retail_Order_ID) are stored directly in the fact table.
9. Only four dimensions are needed, fully covering all entities and relationships from the ER diagram.

Star Schema Relational Diagram



Create Dimension tables

The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the Object Explorer with the 'Retail_DW' database selected. The right pane shows the SQL query editor with the script for creating dimension tables.

```
-- Create Dim_Date
CREATE TABLE Dim_Date (
    Date_SK INT PRIMARY KEY,
    FullDate DATE NOT NULL,
    Year INT NOT NULL,
    Quarter INT NOT NULL,
    Month INT NOT NULL,
    MonthName VARCHAR(20) NOT NULL,
    Day INT NOT NULL,
    WeekdayName VARCHAR(20) NOT NULL
);

-- Populate Dim_Date (2014-2018 - covers your dataset + buffer)
DECLARE @StartDate DATE = '2014-01-01';
DECLARE @EndDate DATE = '2018-12-31';

WHILE @StartDate <= @EndDate
BEGIN
    INSERT INTO Dim_Date
    VALUES (
        YEAR(@StartDate)*10000 + MONTH(@StartDate)*100 + DAY(@StartDate),
        @StartDate,
        YEAR(@StartDate),
        DATEPART(QUARTER, @StartDate),
        MONTH(@StartDate),
        DATENAME(MONTH, @StartDate),
        DAY(@StartDate),
        DATENAME(WEEKDAY, @StartDate)
    );
    SET @StartDate = DATEADD(DAY, 1, @StartDate);
END
```

The script continues with the creation of other dimension tables:

```
CREATE TABLE Dim_SalesPerson (
    SalesPerson_SK INT IDENTITY(1,1) PRIMARY KEY,
    Retail_Sales_People VARCHAR(100) NOT NULL
);

CREATE TABLE Dim_Product (
    Product_SK INT IDENTITY(1,1) PRIMARY KEY,
    Product_ID VARCHAR(30) NOT NULL,
    Product_Name VARCHAR(200) NOT NULL,
    Sub_Category VARCHAR(100) NOT NULL,
    Category VARCHAR(50) NOT NULL,
    Unit_CP DECIMAL(18,4) NOT NULL,
    Unit_SP DECIMAL(18,4) NOT NULL
);

CREATE TABLE Dim_Customer (
    Customer_SK INT IDENTITY(1,1) PRIMARY KEY,
    Customer_ID VARCHAR(20) NOT NULL,
    Customer_Name VARCHAR(100) NOT NULL,
    Segment VARCHAR(50) NOT NULL,
    Postal_Code VARCHAR(20),
    Country VARCHAR(50),
    City VARCHAR(100),
    State VARCHAR(50),
    Region VARCHAR(50),
    Latitude DECIMAL(10,6),
    Longitude DECIMAL(10,6),
    StartDate DATE NOT NULL,
    EndDate DATE DEFAULT '9999-12-31',
    IsCurrent BIT DEFAULT 1
);
```

Object Explorer

Connect

DESKTOP-3MQSVBK\MSSQLSERVER1 (SQL Ser

Databases

System Databases

Database Snapshots

Retail_OLTP_Source

SLIIT_Retail_DW

SLIIT_Retail_Staging

SLIIT_RetailSourceDB

Retail_DW

Database Diagrams

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.Dim_Customer

dbo.Dim_Date

dbo.Dim_Product

dbo.Dim_SalesPerson

dbo.Fact_OrderLine

Dropped Ledger Tables

Views

External Resources

Synonyms

Programmability

Query Store

Service Broker

Storage

Security

Retail_Staging

Security

Server Objects

Replication

RetrieveStagi...ing (sa (73))

SQLQuery4.s...ot connected*

SQLQueryCre...t connected

SQLQuery1.s...ot connected*

```

67
68
69 CREATE TABLE Fact_OrderLine (
70     OrderLine_SK INT IDENTITY(1,1) PRIMARY KEY,
71     Retail_Order_ID INT NOT NULL,      -- Natural key for Task 6 update
72     Order_ID VARCHAR(30),
73     Date_SK INT NOT NULL,
74     Customer_SK INT NOT NULL,
75     Product_SK INT NOT NULL,
76     SalesPerson_SK INT NOT NULL,
77     Quantity INT NOT NULL,
78     Sales DECIMAL(18,4) NOT NULL,
79     Discount DECIMAL(5,4) NOT NULL,
80     Profit DECIMAL(18,4) NOT NULL,
81     Cost DECIMAL(18,4) NOT NULL,
82     Returned VARCHAR(10),
83     DaysToShip INT,
84     -- Accumulating Fact Columns (Task 6 - 20 marks)
85     accm_txn_create_time DATETIME NULL,
86     accm_txn_complete_time DATETIME NULL,
87     txn_process_time_hours INT NULL,
88
89     FOREIGN KEY (Date_SK) REFERENCES Dim_Date(Date_SK),
90     FOREIGN KEY (Customer_SK) REFERENCES Dim_Customer(Customer_SK),
91     FOREIGN KEY (Product_SK) REFERENCES Dim_Product(Product_SK),
92     FOREIGN KEY (SalesPerson_SK) REFERENCES Dim_SalesPerson(SalesPerson_SK)
93 );

```

100 %

No issues found

Stored Procedures

SQLQuery24...W (sa (65))

SQLQuery23.s...DW (sa (59))*

```

1  USE [Retail_DW]
2  GO
3  /***** Object:  StoredProcedure [dbo].[UpdateDimCustomer]    Script Date: 29/03/2026 05:18:27 *****/
4  SET ANSI_NULLS ON
5  GO
6  SET QUOTED_IDENTIFIER ON
7  GO
8  -- Dim_Customer (SCD Type 2)
9  ALTER PROCEDURE [dbo].[UpdateDimCustomer]
10     @CustomerID      VARCHAR(20),
11     @CustomerName     VARCHAR(50),
12     @Segment          VARCHAR(50),
13     @PostalCode       VARCHAR(50),
14     @Country          VARCHAR(50),
15     @City             VARCHAR(50),
16     @State            VARCHAR(50),
17     @Region           VARCHAR(50),
18     @Latitude         VARCHAR(50),
19     @Longitude        VARCHAR(50)
20 AS
21 BEGIN
22     -- Expire old record if changed
23     UPDATE dc
24     SET EndDate = GETDATE(), IsCurrent = 0
25     FROM dbo.Dim_Customer dc
26     WHERE dc.Customer_ID = @CustomerID
27     AND dc.IsCurrent = 1
28     AND (dc.Customer_Name <> @CustomerName OR dc.City <> @City OR dc.State <> @State);
29
30     -- Insert new record (or changed version)
31     IF NOT EXISTS (
32         SELECT 1 FROM dbo.Dim_Customer
33         WHERE Customer_ID = @CustomerID AND IsCurrent = 1
34     )
35     BEGIN
36         INSERT INTO dbo.Dim_Customer (
37             Customer_ID, Customer_Name, Segment, Postal_Code, Country,
38             City, State, Region, Latitude, Longitude,
39             StartDate, EndDate, IsCurrent
40         )
41         VALUES (
42             @CustomerID, @CustomerName, @Segment, @PostalCode, @Country,
43             @City, @State, @Region, @Latitude, @Longitude,
44             GETDATE(), '9999-12-31', 1
45         );
46     END
47 END;
48

```

75 %

Connected. (1/1)

SQLQuery25...W (sa (68))

SQLQuery23.s...DW (sa (59))*

```

1  USE [Retail_DW]
2  GO
3  /***** Object:  StoredProcedure [dbo].[UpdateDimProduct]    Script Date: 29/03/2026 05:19:42 *****/
4  SET ANSI_NULLS ON
5  GO
6  SET QUOTED_IDENTIFIER ON
7  GO
8  -- Dim_Product Load/Update
9  ALTER PROCEDURE [dbo].[UpdateDimProduct]
10     @ProductID      NVARCHAR(255),
11     @ProductName     NVARCHAR(255),
12     @SubCategory     NVARCHAR(255),
13     @Category        NVARCHAR(255),
14     @UnitCP          FLOAT,
15     @UnitSP          FLOAT
16 AS
17 BEGIN
18     IF NOT EXISTS (SELECT 1 FROM dbo.Dim_Product WHERE Product_ID = @ProductID)
19     BEGIN
20         INSERT INTO dbo.Dim_Product
21         (Product_ID, Product_Name, Sub_Category, Category, Unit_CP, Unit_SP)
22         VALUES
23         (@ProductID, @ProductName, @SubCategory, @Category, @UnitCP, @UnitSP);
24     END
25     ELSE
26     BEGIN
27         UPDATE dbo.Dim_Product
28         SET Product_Name = @ProductName,
29             Sub_Category = @SubCategory,
30             Category = @Category,
31             Unit_CP = parameter @Category nvarchar(255),
32             Unit_SP = parameter @Category nvarchar(255)
33         WHERE Product_ID = @ProductID;
34     END
35 END;
36

```

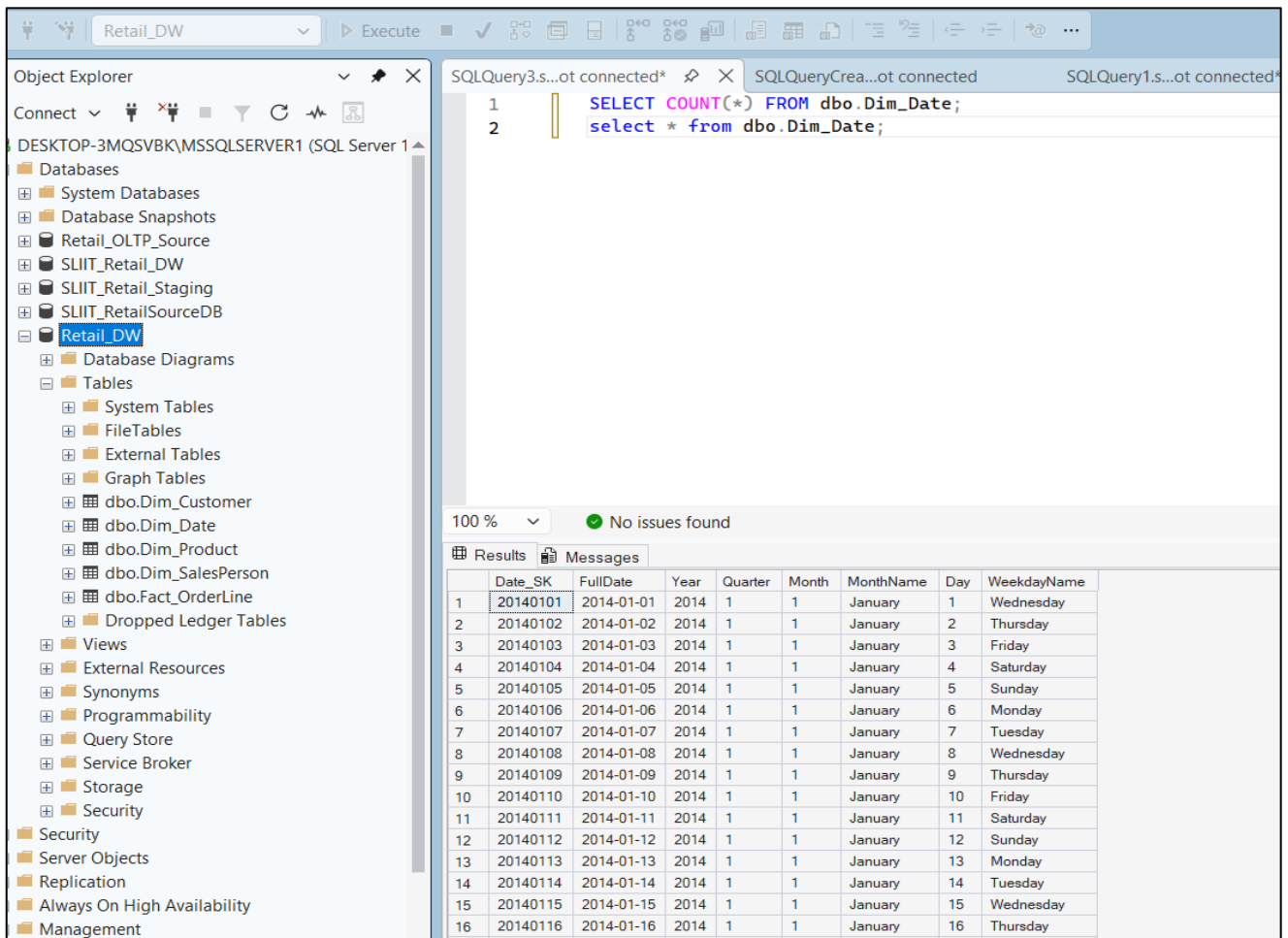
75 %

Connected. (1/1)

```

1  USE [Retail_Dw]
2  GO
3  /***** Object: StoredProcedure [dbo].[UpdateDimSalesPerson]    Script Date: 29/03/2026 05:28:50 *****/
4  SET ANSI_NULLS ON
5  GO
6  SET QUOTED_IDENTIFIER ON
7  GO
8  -- Dim_SalesPerson Load/Update
9  ALTER PROCEDURE [dbo].[UpdateDimSalesPerson]
10     @RetailSalesPeople NVARCHAR(200)
11  AS
12  BEGIN
13      IF NOT EXISTS (SELECT 1 FROM dbo.Dim_SalesPerson WHERE Retail_Sales_People = @RetailSalesPeople)
14      BEGIN
15          INSERT INTO dbo.Dim_SalesPerson (Retail_Sales_People)
16          VALUES (@RetailSalesPeople);
17      END
18      -- No UPDATE needed for SalesPerson (no changing attributes)
19  END;
20

```



Task 5: ETL Development

I designed and executed an ETL (Extract, Transform, Load) workflow to move and prepare data from multiple sources into the data warehouse's fact table (Fact_OrderLine). The process was implemented in SSIS and followed best practices for data warehousing.

Order of Execution and Task Details

1. Data Extraction

- Sources Used:
 - A SQL Server staging table (Stg_OrderLines) containing all order and order-line transactional data.
 - An Excel spreadsheet (product_master.xlsx) containing product master data.
 - A CSV file (customer_master.csv) containing customer master data.
- Action:
 - Configured OLE DB Source (Staging table), Excel Source (product_master.xlsx), and Flat File Source (customer_master.csv) components in SSIS to extract data from all three heterogeneous sources.

2. Data Transformation

- Lookups:
 - Used Lookup transformations to resolve surrogate keys for all dimensions (Date_SK, Customer_SK, Product_SK, SalesPerson_SK) by matching business keys from the staging data with the dimension tables.

- Data Conversion:
 - Applied Data Conversion transformation to ensure data types (e.g., Order_Date → Date_SK, Returned → ReturnedFlag) matched the destination schema.
- Derived Columns:
 - Added Derived Column transformations to create business calculated columns: DiscountAmount = Sales × Discount, ReturnedFlag, and DaysToShip = DATEDIFF(day, Order_Date, Ship_Date).
- Conditional Split:
 - Implemented a Conditional Split to demonstrate ETL logic by routing rows based on Sales amount (e.g., splitting into “HighValueOrder” where Sales > 500 and “NormalOrder” paths).
- Slowly Changing Dimension:
 - Used Slowly Changing Dimension (SCD) Wizard (Type 2) on the Customer source to handle historical changes in region, segment, and address.
- Incremental Load Logic:
 - Added a Lookup transformation before loading to the fact table to check for existing Retail_Order_ID values, ensuring only new records are inserted and avoiding primary key violations.

3. Data Loading

- Destination:
 - Loaded the transformed data into the Fact_OrderLine table in the data warehouse using OLE DB Destination.
- Incremental Load:
 - Only rows that did not already exist in the fact table (as determined by the Lookup) were inserted, ensuring no duplicates.

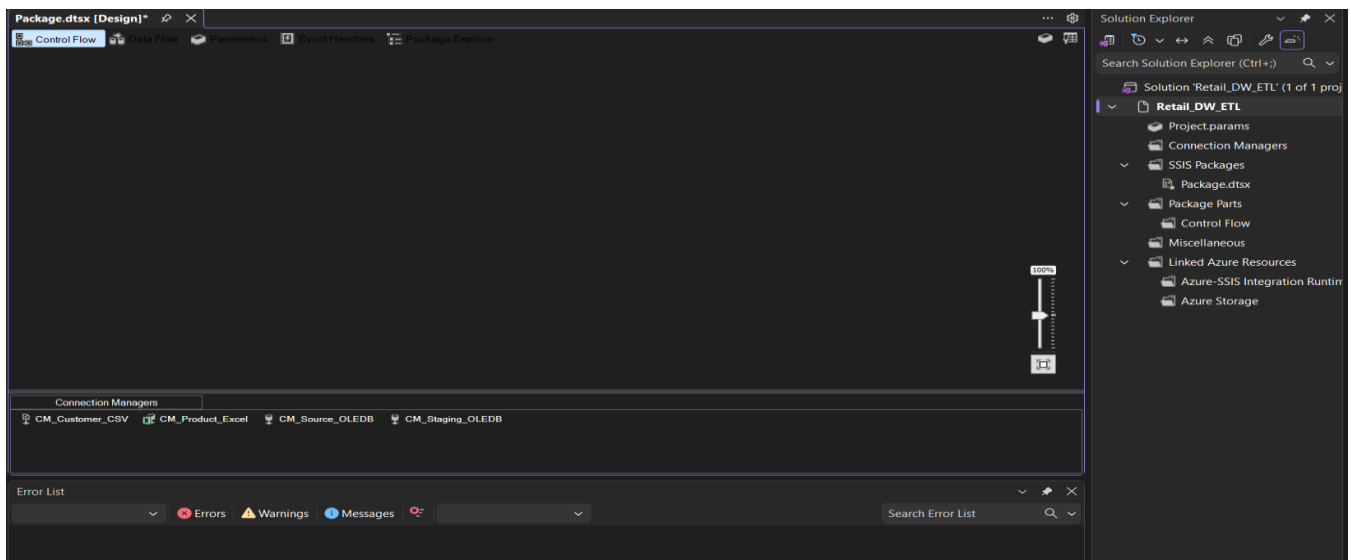
4. Error Handling

- Warnings and Errors:
 - Configured SSIS to redirect or log rows that failed due to data type mismatches, lookup failures, or constraint violations.
- Validation:
 - Used Data Viewers during development to confirm that the correct rows were flowing through each transformation and into the destination.

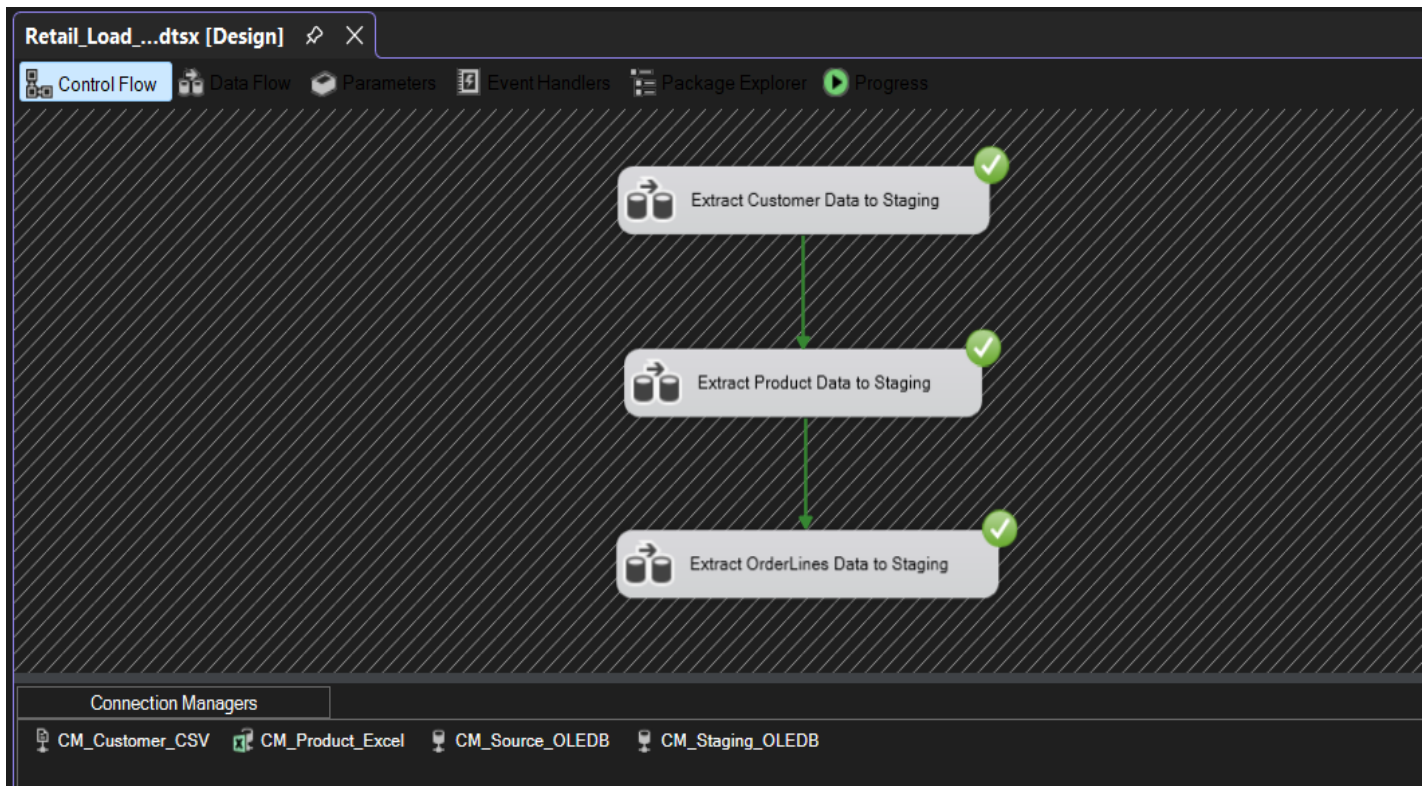
Execution Order Summery

1. Extract data from SQL Server staging table, Excel, and CSV sources into the SSIS data flow.
2. Perform Lookups to resolve dimension surrogate keys.
3. Apply Data Conversion and Derived Column transformations.
4. Use SCD Type 2 Wizard for Dim_Customer.
5. Use Conditional Split to demonstrate business logic.
6. Use a final Lookup to filter out duplicates (incremental load).
7. Load the cleaned and transformed data into the Fact_OrderLine table and handle/log any errors or warnings encountered.

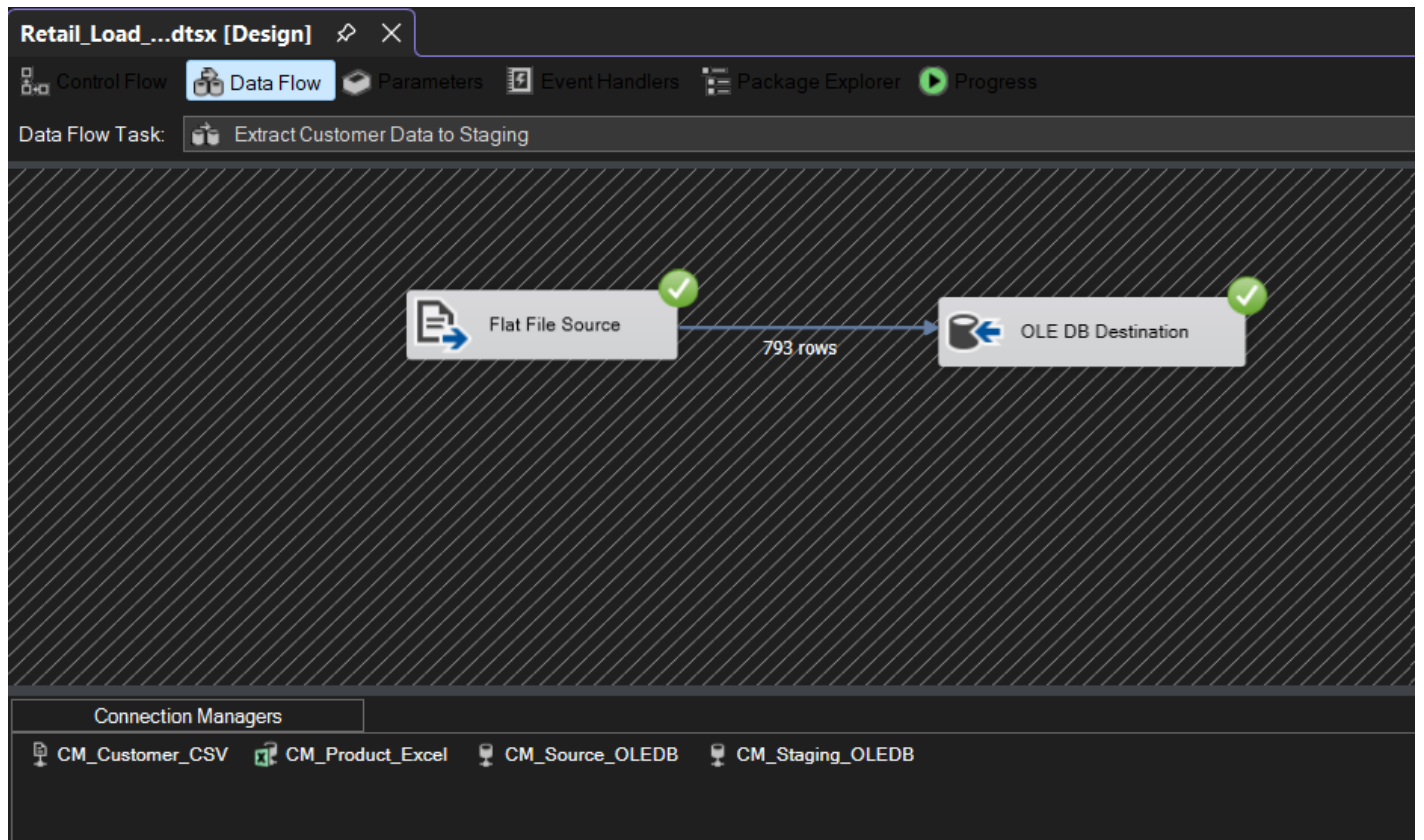
Create Retail_DW_ETL and Connection managers



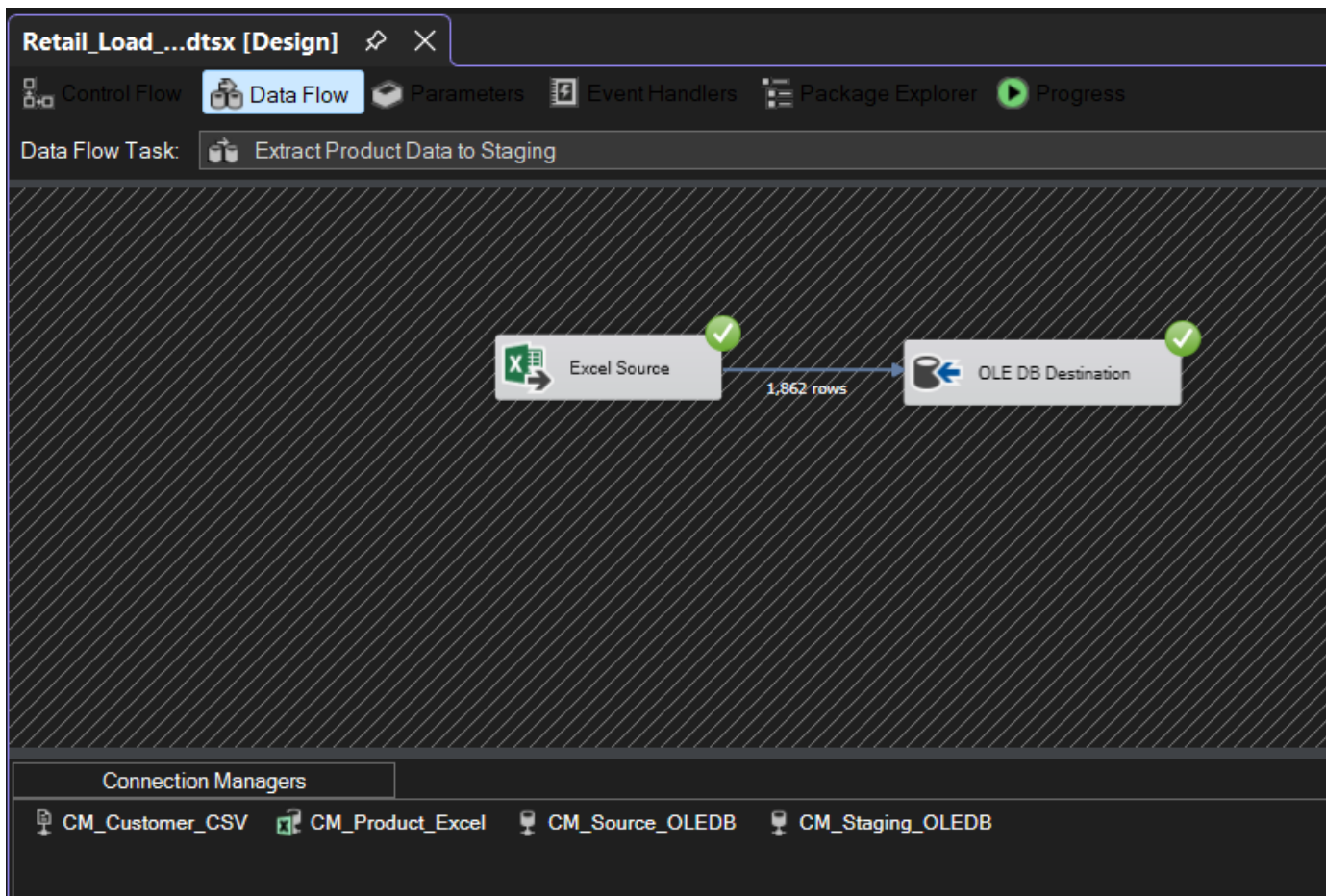
ETL – Source to Staging



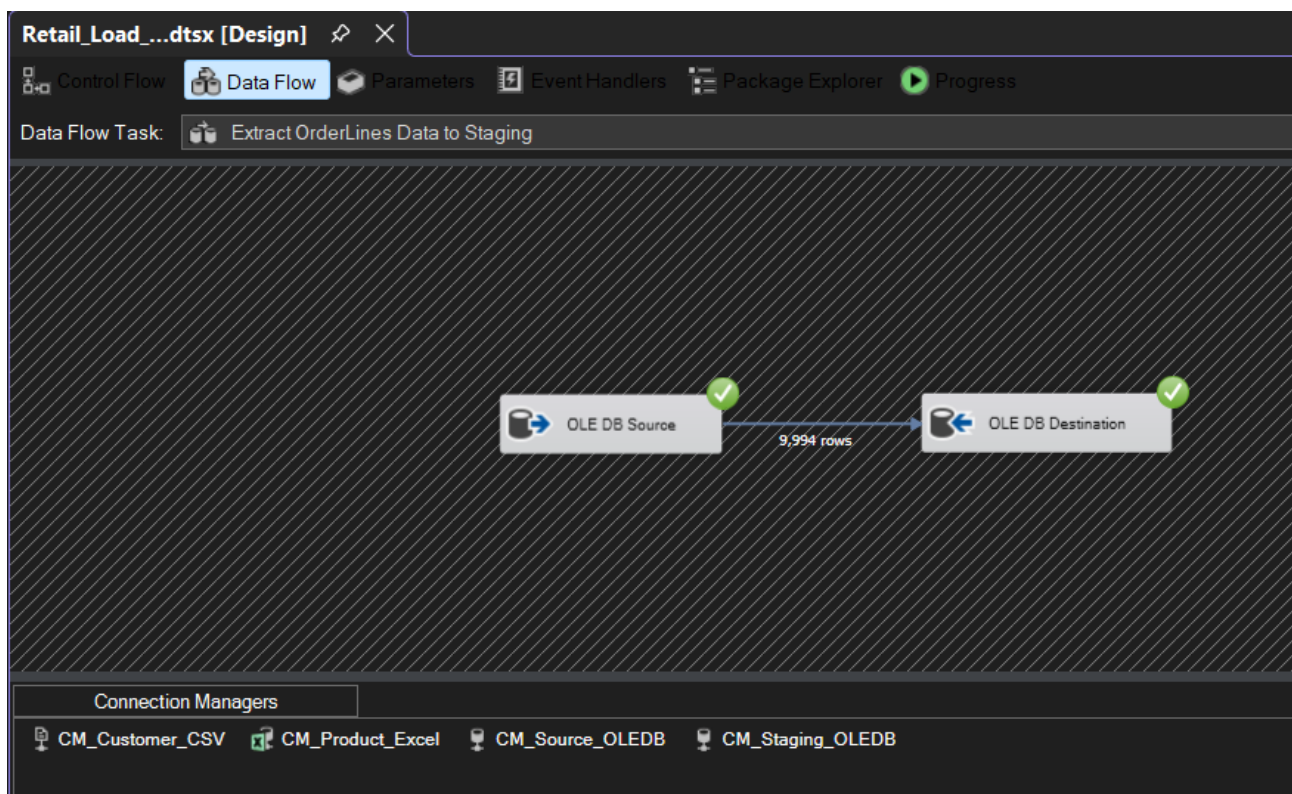
Extract Customer data to staging



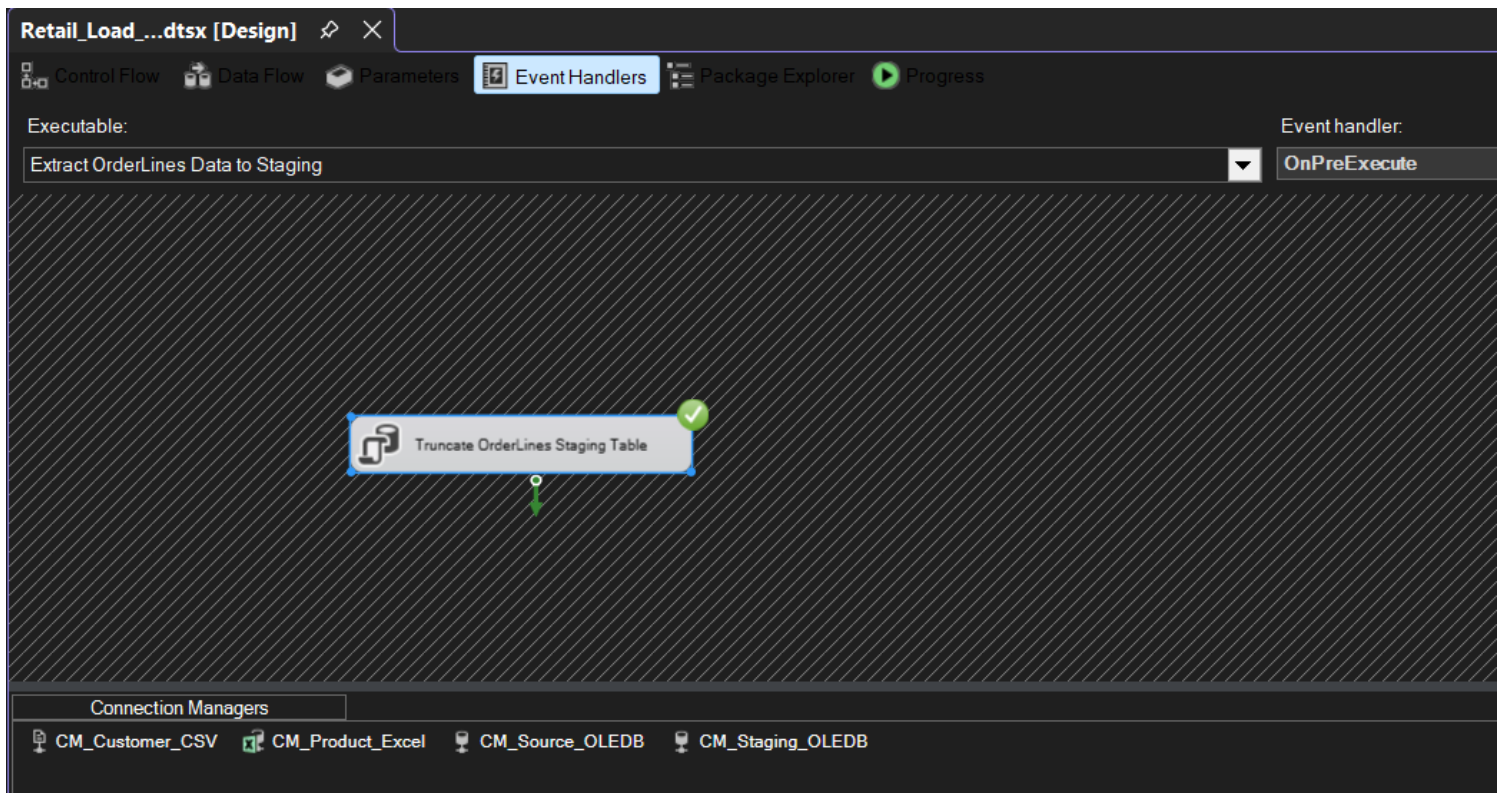
Extract Product data to staging



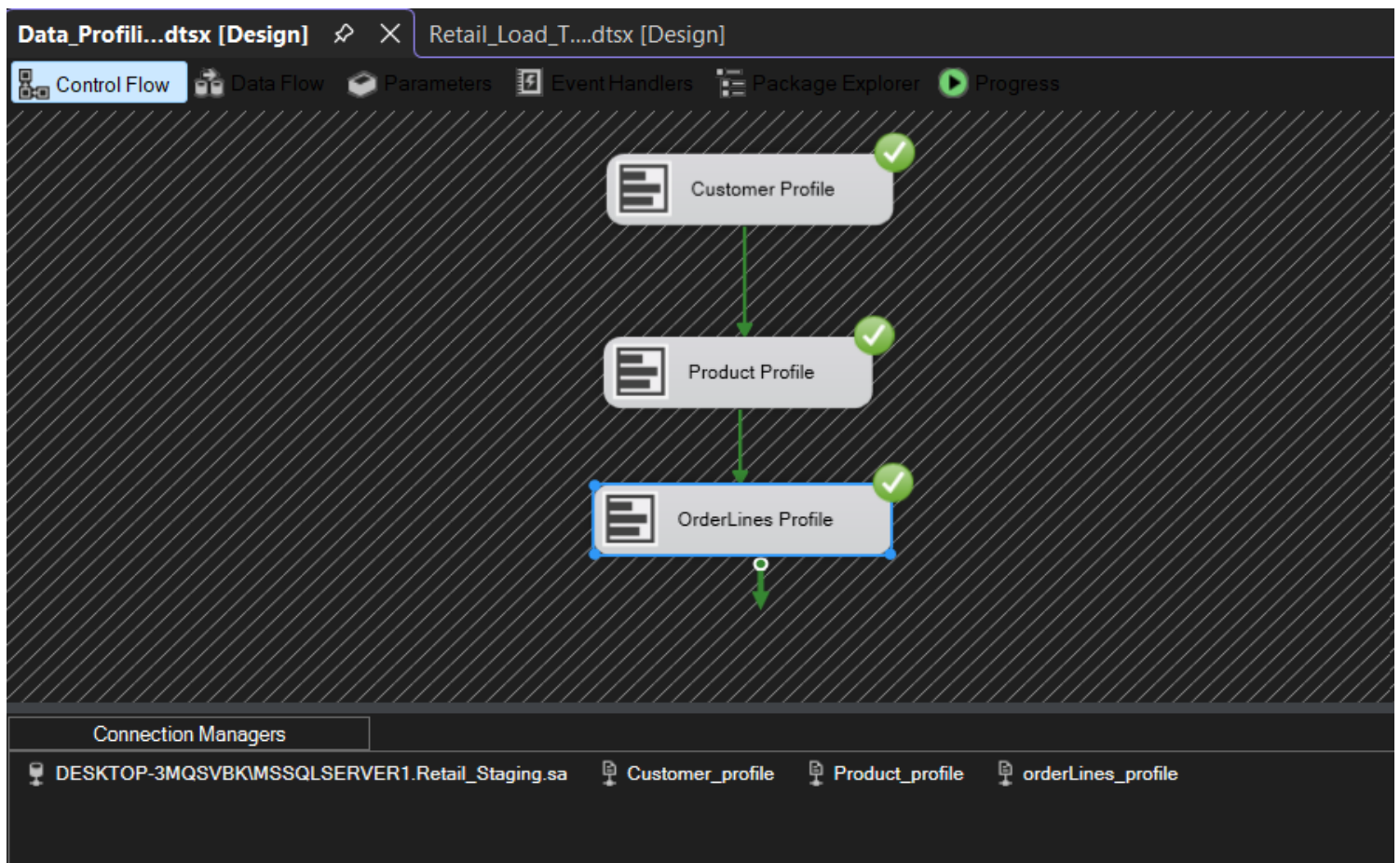
Extract OrderLines data to staging



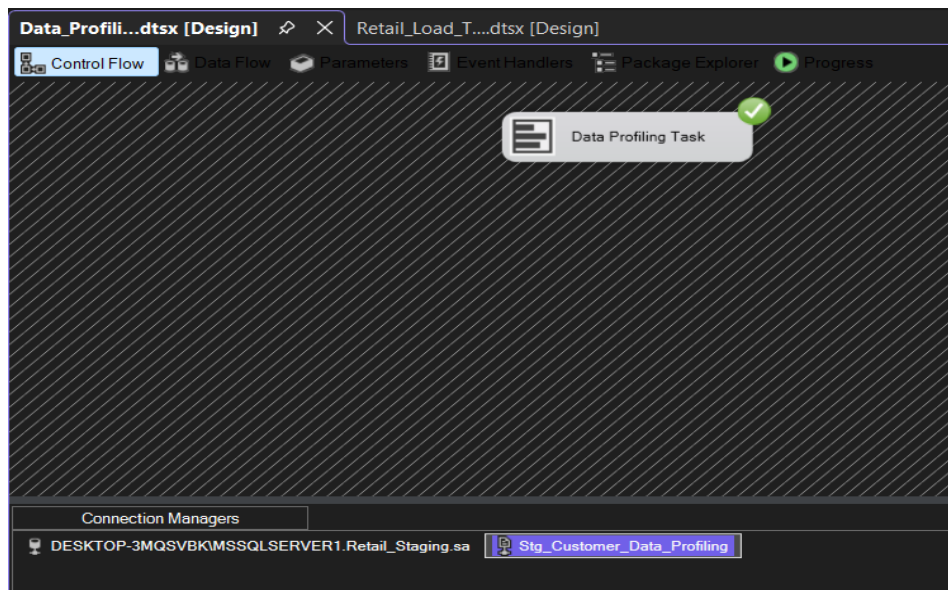
Truncate OrderLines Staging table



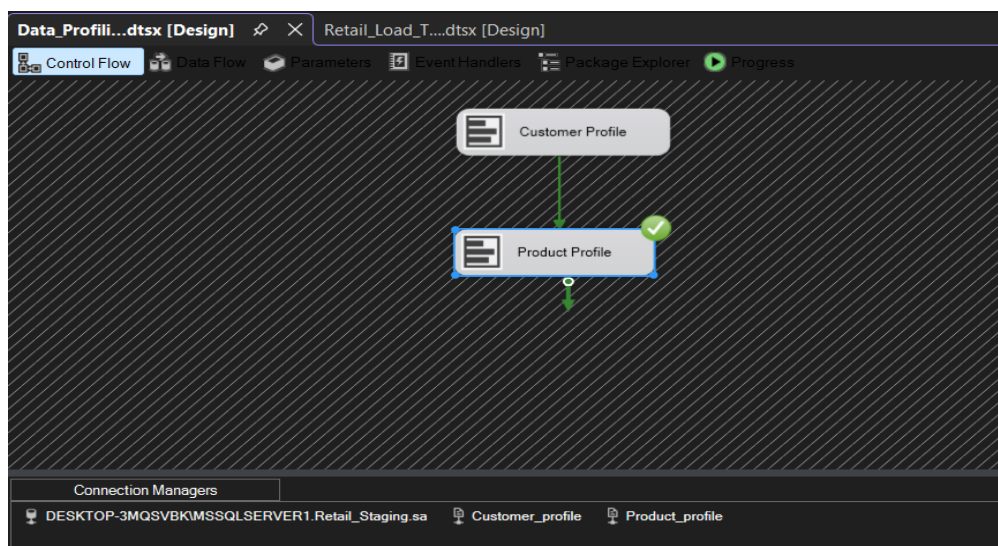
ETL - Data Profiling



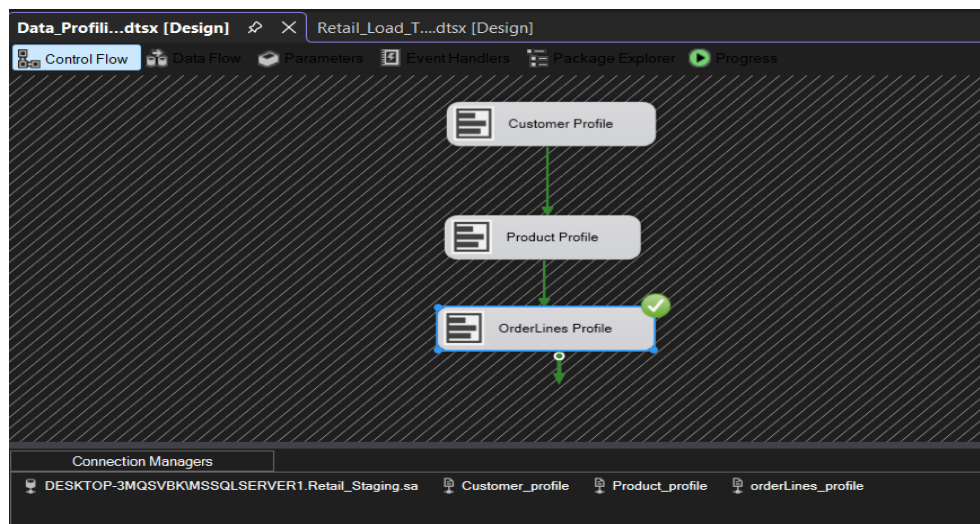
Customer Data Profile



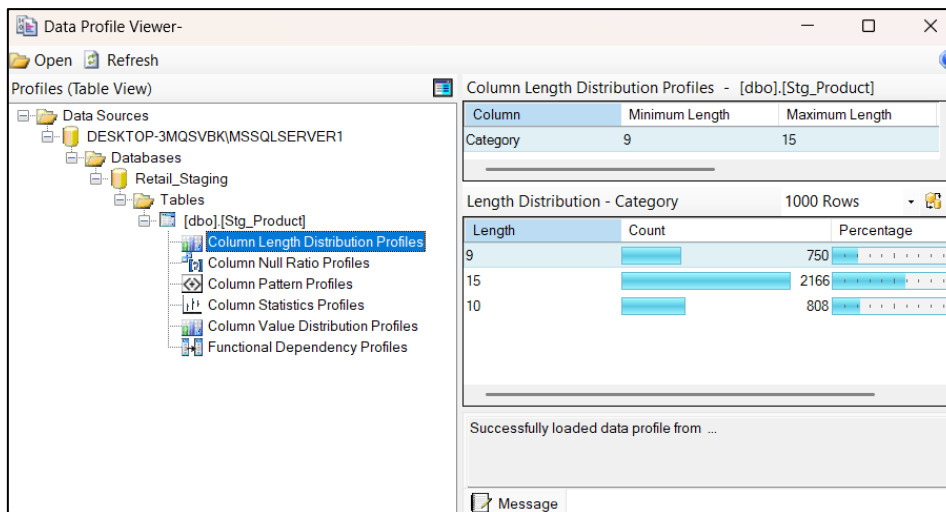
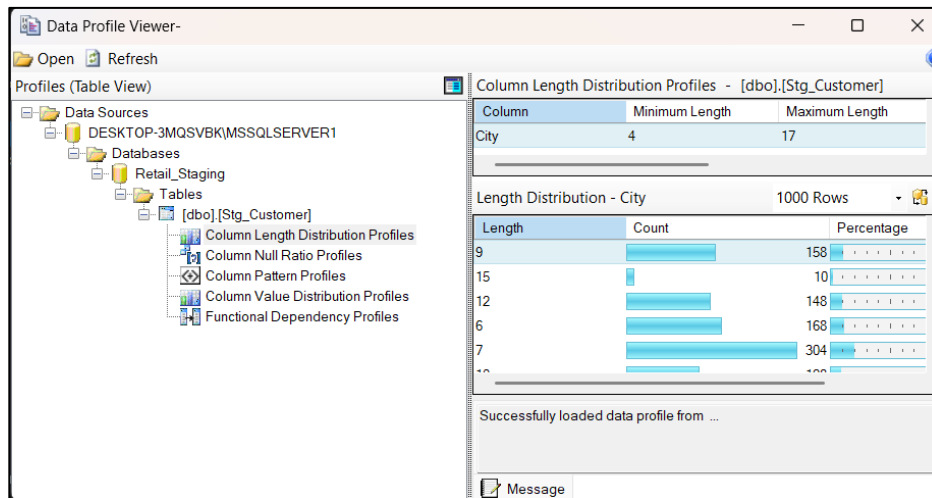
Product Data Profile



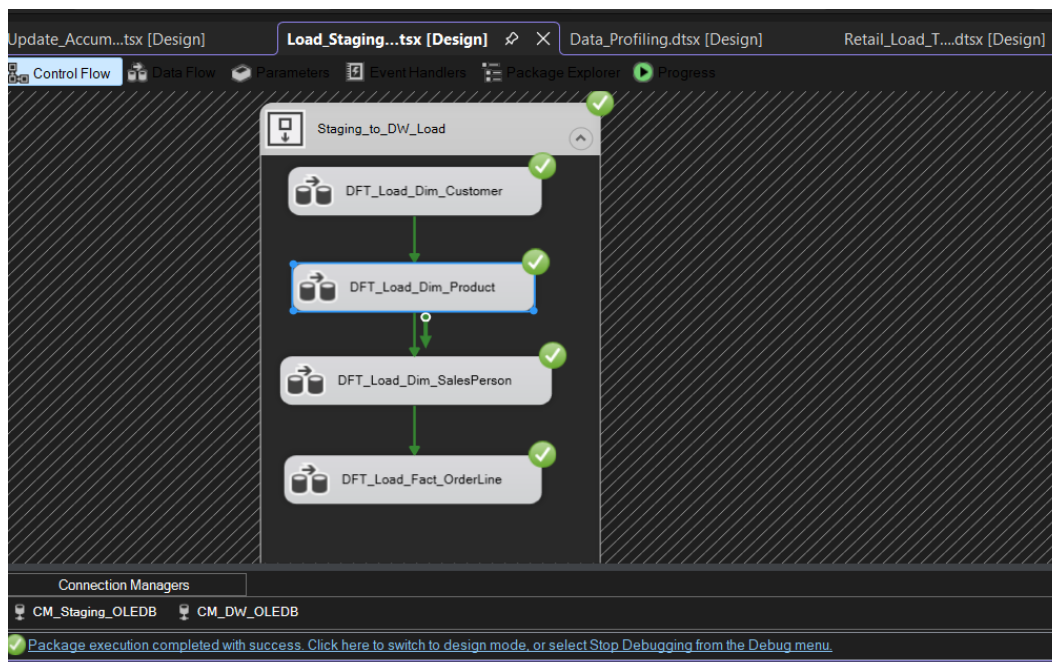
OrderLines Data Profile



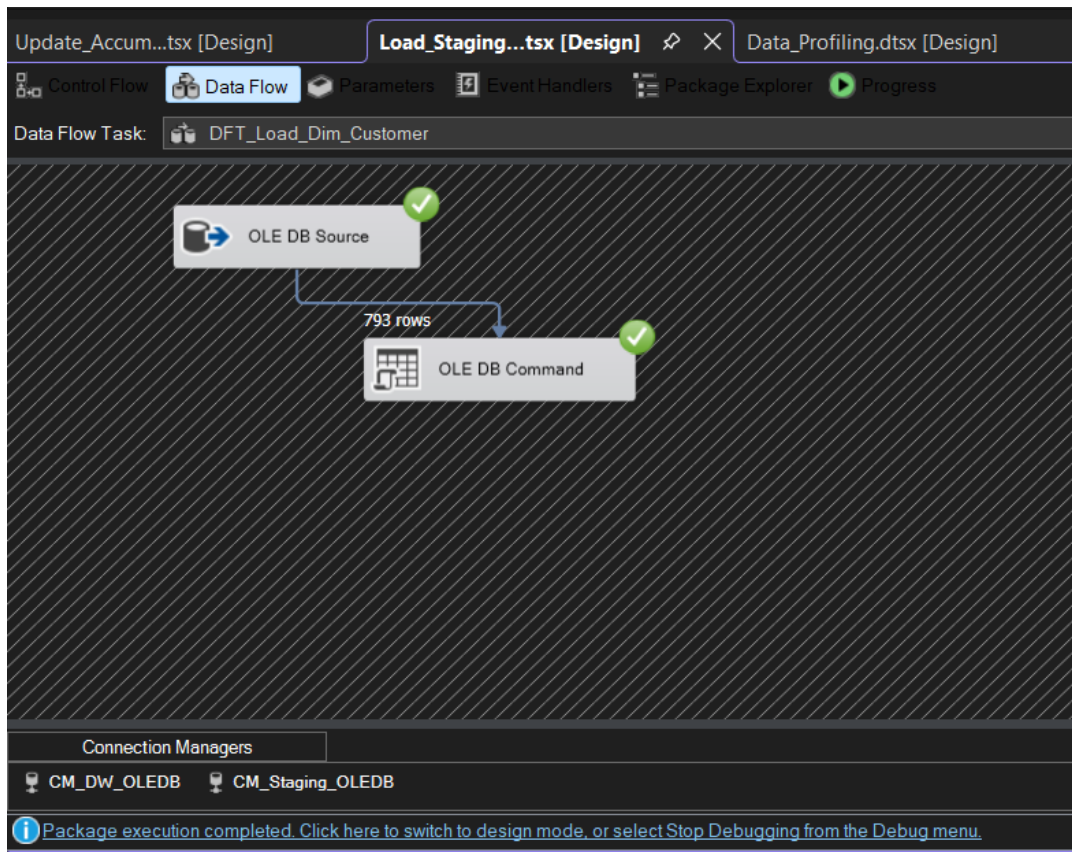
Data Profile Viewer



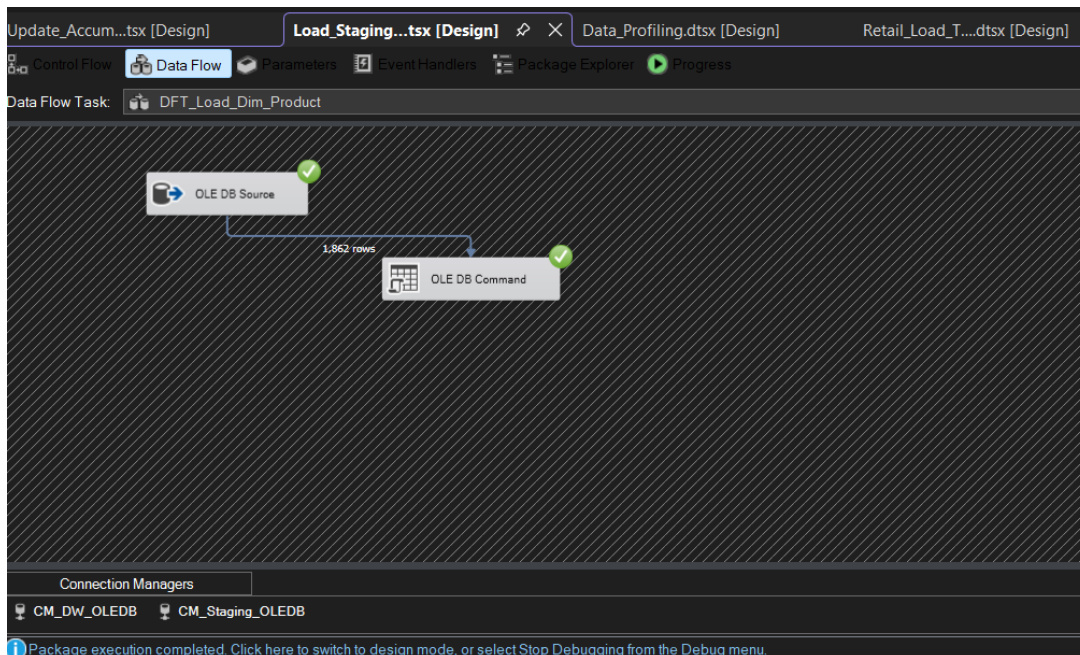
ETL System to Datawarehouse



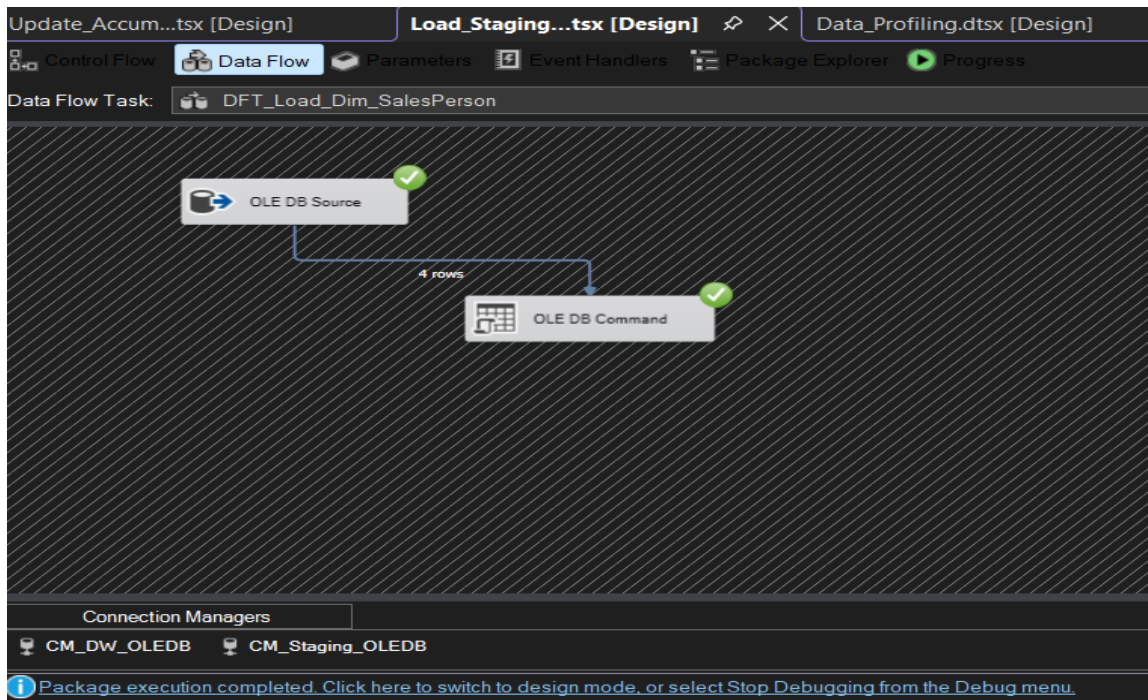
Transfer and Load Dim_Customer Data from staging



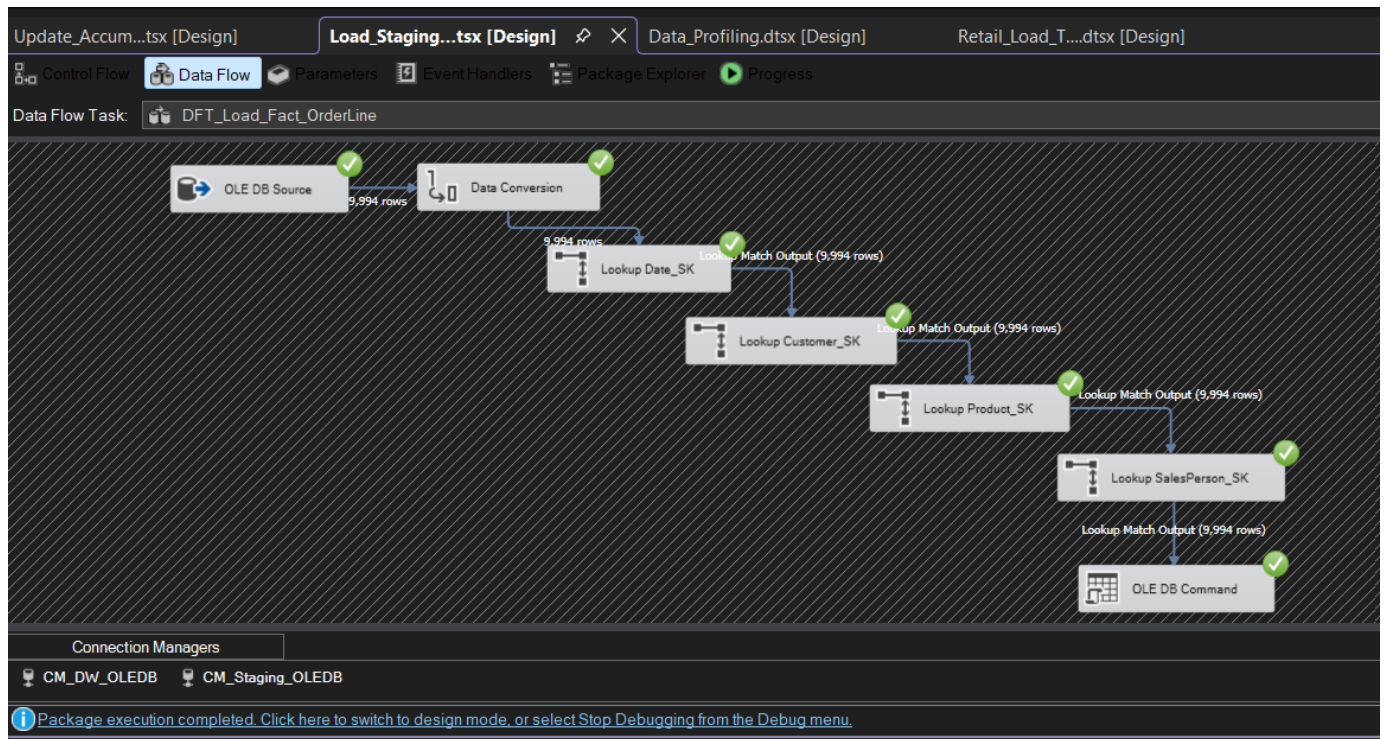
Transfer and Load Dim_Product Data from staging



Transfer and Load Dim_SalesPerson Data from staging



Transfer and Load Fact_OrderLine Data from staging



Retrieve Data from Dim_Customer table

SQLQuery29....W (sa (66))* X SQLQuery23.s...DW (sa (59))*

1 `select * from dbo.Dim_Customer;`

110 % No issues found Ln: 1, Ch: 1 (31 chars, 1 lines) TABS CRLF Windows 1252

	Customer_SK	Customer_ID	Customer_Name	Segment	Postal_Code	Country	City	State	Region	Latitude	Longitude	StartDate	EndDate	IsCurrent
1	1	CG-12520	Claire Gute	Consumer	42420	United States	Henderson	Kentucky	South	37.839333	-84.270018	2026-03-28	9999-12-31	1
2	2	DV-13045	Darrin Van Huff	Corporate	90036	United States	Los Angeles	California	West	36.778261	-119.417932	2026-03-28	9999-12-31	1
3	3	SO-20335	Sean O'Donnell	Consumer	33311	United States	Fort Lauderdale	Florida	South	27.664827	-81.515754	2026-03-28	9999-12-31	1
4	4	BH-11710	Brosina Hoffman	Consumer	90032	United States	Los Angeles	California	West	36.778261	-119.417932	2026-03-28	9999-12-31	1
5	5	AA-10480	Andrew Allen	Consumer	28027	United States	Concord	North Carolina	South	35.759573	-79.0193	2026-03-28	9999-12-31	1
6	6	IM-15070	Irene Maddox	Consumer	98103	United States	Seattle	Washington	West	47.751074	-120.740139	2026-03-28	9999-12-31	1
7	7	HP-14815	Harold Pawlan	Home Office	76106	United States	Fort Worth	Texas	Central	31.968599	-99.901813	2026-03-28	9999-12-31	1
8	8	PK-19075	Pete Kriz	Consumer	53711	United States	Madison	Wisconsin	Central	43.78444	-88.787868	2026-03-28	9999-12-31	1
9	9	AQ-10270	Alejandro Grove	Consumer	84084	United States	West Jordan	Utah	West	39.32098	-111.093731	2026-03-28	9999-12-31	1
10	10	ZD-21925	Zuschuss Donatelli	Consumer	94109	United States	San Francisco	California	West	36.778261	-119.417932	2026-03-28	9999-12-31	1
11	11	KB-16585	Ken Black	Corporate	68025	United States	Fremont	Nebraska	Central	41.492537	-99.901813	2026-03-28	9999-12-31	1
12	12	SF-20065	Sandra Flanagan	Consumer	19140	United States	Philadelphia	Pennsylvania	East	41.203322	-77.194525	2026-03-28	9999-12-31	1
13	13	EB-13870	Emily Burns	Consumer	84057	United States	Orem	Utah	West	39.32098	-111.093731	2026-03-28	9999-12-31	1
14	14	EH-13945	Eric Hoffmann	Consumer	90049	United States	Los Angeles	California	West	36.778261	-119.417932	2026-03-28	9999-12-31	1
15	15	TB-21520	Tracy Blumstein	Consumer	19140	United States	Philadelphia	Pennsylvania	East	41.203322	-77.194525	2026-03-28	9999-12-31	1
16	16	MA-17560	Matt Abelman	Home Office	77095	United States	Houston	Texas	Central	31.968599	-99.901813	2026-03-28	9999-12-31	1
17	17	GL-14485	Gene Hale	Corporate	75080	United States	Richardson	Texas	Central	31.968599	-99.901813	2026-03-28	9999-12-31	1

Query executed successfully. DESKTOP-3MQSVBK\MSSQLSERVER... sa (66) Retail_DW 00:00:00 Row: 1, Col: 1 793 rows

Retrieve Data from Dim_Product table

SQLQuery29....W (sa (66))* X SQLQuery23.s...DW (sa (59))*

1 `select * from dbo.Dim_Product;`

110 % No issues found Ln: 1, Ch: 1 (30 chars, 1 lines) TABS CRLF Windows 1252

	Product_SK	Product_ID	Product_Name	Sub_Category	Category	Unit_CP	Unit_SP
1	1	FUR-BO-10001798	Bush Somerset Collection Bookcase	Bookcases	Furniture	110.02319999999999	130.98
2	2	FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Ro...	Chairs	Furniture	170.786	243.98
3	3	OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters by Univ...	Labels	Office Supplies	3.8742999999999999	7.31
4	4	FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	Tables	Furniture	268.1217	191.5155
5	5	OFF-ST-10000760	Eldon Fold 'N Roll Cart System	Storage	Office Supplies	9.9257999999999999	11.184
6	6	FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessor...	Furnishings	Furniture	4.9558	6.9799999999999999
7	7	OFF-AR-10002833	Newell 322	Art	Office Supplies	1.3286	1.82
8	8	TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	Phones	Technology	136.0728	151.192
9	9	OFF-BI-10003910	DXL Angle-View Binders with Locking Rings by Sam...	Binders	Office Supplies	4.2405	6.168
10	10	OFF-AP-10002892	Belkin F5C206/TEL 6 Outlet Surge	Appliances	Office Supplies	16.086	22.98
11	11	FUR-TA-10001539	Chromcraft Rectangular Conference Tables	Tables	Furniture	180.0972	189.576
12	12	TEC-PH-10002033	Konftel 250 Conference phone - Charcoal black	Phones	Technology	210.7668	227.856
13	13	OFF-PA-10002365	Xerox 1967	Paper	Office Supplies	3.3695999999999999	5.184
14	14	OFF-BI-10003656	Fellowes PB200 Plastic Comb Binding Machine	Binders	Office Supplies	91.7946	135.992
15	15	OFF-AP-10002311	Holmes Replacement Filter for HEPA Air Cleaner, Ve...	Appliances	Office Supplies	35.5336	13.762
16	16	OFF-BI-10000756	Storax DuraTech Recycled Plastic Frosted Binders	Binders	Office Supplies	2.1199999999999999	0.848
17	17	OFF-ST-10004188	Storax DuraTech Shakers Vertical R-Shelf 7234 x 3635 x	Storage	Office Supplies	108.7604	110.88

Query executed successfully. DESKTOP-3MQSVBK\MSSQLSERVER... sa (66) Retail_DW 00:00:00 Row: 1, Col: 1 1,862 rows

Retrieve Data from Dim_Product table

SQLQuery29....W (sa (66))* X SQLQuery23.s...DW (sa (59))*

```
1 1 select * from dbo.Dim_SalesPerson;
```

110 % No issues found Ln: 1, Ch: 1 (34 chars, 1 lines) TABS CRLF Windows 1252

Results Messages

	SalesPerson_SK	Retail_Sales_People
1	1	Anna Andreadi
2	2	Cassandra Brandow
3	3	Kelly Williams
4	4	Chuck Magee

Query executed successfully. DESKTOP-3MQSVBK\MSSQLSERVER... sa (66) Retail_DW 00:00:00 Row: 1, Col: 1 4 rows

Retrieve Data from Fact_OrderLine table

SQLQuery29....W (sa (66))* X SQLQuery23.s...DW (sa (59))*

```
1 1 select * from dbo.Fact_OrderLine;
```

110 % No issues found Ln: 1, Ch: 1 (33 chars, 1 lines) TABS CRLF Windows 1252

Results Messages

	OrderLine_SK	Retail_Order_ID	Order_ID	Date_SK	Customer_SK	Product_SK	SalesPerson_SK	Quantity	Sales	Discount	Profit	Cost	Returned	DaysToShip	accn_txn_create_t
1	1	1027	CA-2016-126529	20161101	369	780	4	3	63.3120000000	0.2000000000	20.5764000000	42.7356000000	Yes	0	2026-03-29 01:49:!
2	2	1029	CA-2014-163552	20141107	157	259	4	5	177.2000000000	0.0000000000	83.2840000000	93.9160000000	Not	0	2026-03-29 01:49:!
3	3	1034	CA-2016-109820	20161120	370	784	4	3	15.6960000000	0.2000000000	5.1012000000	10.5948000000	Not	2	2026-03-29 01:49:!
4	4	1037	CA-2016-113061	20160422	331	785	3	2	86.6200000000	0.0000000000	8.6620000000	77.9580000000	Not	4	2026-03-29 01:49:!
5	5	1038	CA-2015-127418	20150613	371	786	1	3	36.6240000000	0.2000000000	13.7340000000	22.8900000000	Not	1	2026-03-29 01:49:!
6	6	1039	CA-2017-121818	20171120	372	787	2	7	23.9680000000	0.2000000000	2.6964000000	21.2716000000	Not	1	2026-03-29 01:49:!
7	7	1041	CA-2016-127670	20160320	373	617	3	4	697.1600000000	0.0000000000	146.4036000000	550.7564000000	Not	4	2026-03-29 01:49:!
8	8	1042	CA-2016-102981	20160609	374	789	4	2	31.8600000000	0.0000000000	11.1510000000	20.7090000000	Not	92	2026-03-29 01:49:!
9	9	1044	CA-2017-115651	20170907	375	791	3	5	8.0400000000	0.2000000000	2.9835000000	5.8565000000	Not	91	2026-03-29 01:49:!
10	10	1046	CA-2017-152702	20171210	18	793	3	14	254.6040000000	0.3000000000	-18.1860000000	272.7900000000	Not	0	2026-03-29 01:49:!
11	11	1047	CA-2016-169103	20160803	234	639	2	5	1363.9600000000	0.2000000000	85.2475000000	1278.7125000000	Not	0	2026-03-29 01:49:!
12	12	1049	CA-2014-139192	20140527	63	40	1	12	1113.5040000000	0.2000000000	125.2692000000	988.2348000000	Not	0	2026-03-29 01:49:!
13	13	1051	US-2015-153500	20150307	376	794	4	2	168.4640000000	0.2000000000	-29.4812000000	197.9452000000	Not	61	2026-03-29 01:49:!
14	14	1054	CA-2015-110667	20150404	377	797	4	2	11.1600000000	0.0000000000	4.3524000000	6.8076000000	Not	122	2026-03-29 01:49:!
15	15	1058	CA-2017-167150	20171012	131	800	1	4	19.9360000000	0.2000000000	7.2268000000	12.7092000000	Not	66	2026-03-29 01:49:!
16	16	1060	CA-2016-105284	20161124	378	802	4	2	4.4160000000	0.2000000000	1.6080000000	3.8152000000	Not	0	2026-03-29 01:49:!

Query executed successfully. DESKTOP-3MQSVBK\MSSQLSERVER... sa (66) Retail DW 00:00:00 Row: 1, Col: 1 5,009 rows

Task 6: ETL development – Accumulating fact tables

I updated the DiscountAmount and DaysToShip fields and calculated the return flag in the Fact_OrderLine fact table, reflecting the actual discount value, shipping performance, and return status for each order line transaction.

Process Overview

1. Preparation of Calculated Measures Data

A separate dataset (CalculatedMeasures.csv) was prepared, containing:

- Retail_Order_ID (the natural key)
- DiscountAmount (calculated as Sales × Discount)
- DaysToShip (calculated as the difference between Ship_Date and Order_Date)
- ReturnedFlag (converted from “Yes”/“Not” to 1/0)

This file ensures that each order line in the fact table can be matched and updated with its actual calculated values.

2. Design of the Update Package in SSIS

- A new SSIS package (Update_Fact_CalculatedMeasures.dtsx) was created specifically for this update process.
- The package performs the following steps:
 - **Flat File Source:** Reads data from CalculatedMeasures.csv.
 - **Lookup Transformation:** Matches records with the Fact_OrderLine table using Retail_Order_ID.
 - **OLE DB Command:** Executes an UPDATE statement on the Fact_OrderLine table, setting the DiscountAmount, DaysToShip, and ReturnedFlag for each matching Retail_Order_ID.

3. Execution and Validation

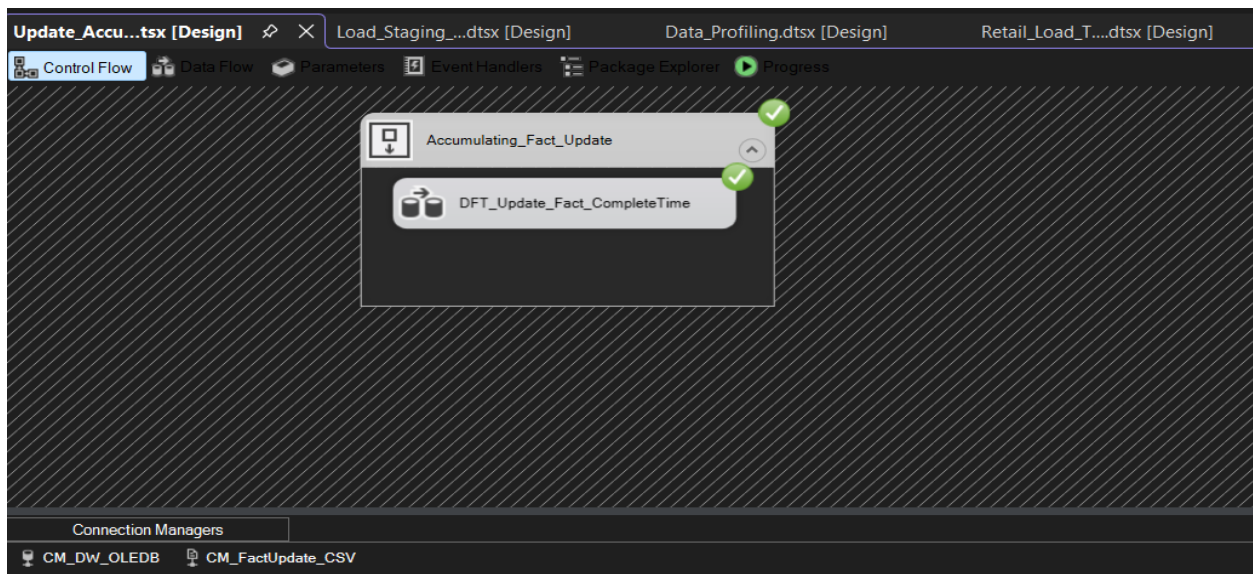
- After execution, the fact table was checked to ensure:
 - DiscountAmount is populated for all order lines.
 - DaysToShip is a positive integer, reflecting the actual shipping performance.
 - ReturnedFlag is correctly set as 0 or 1 for return analysis.
- Data Viewers and SQL queries were used to validate the updates.

Result

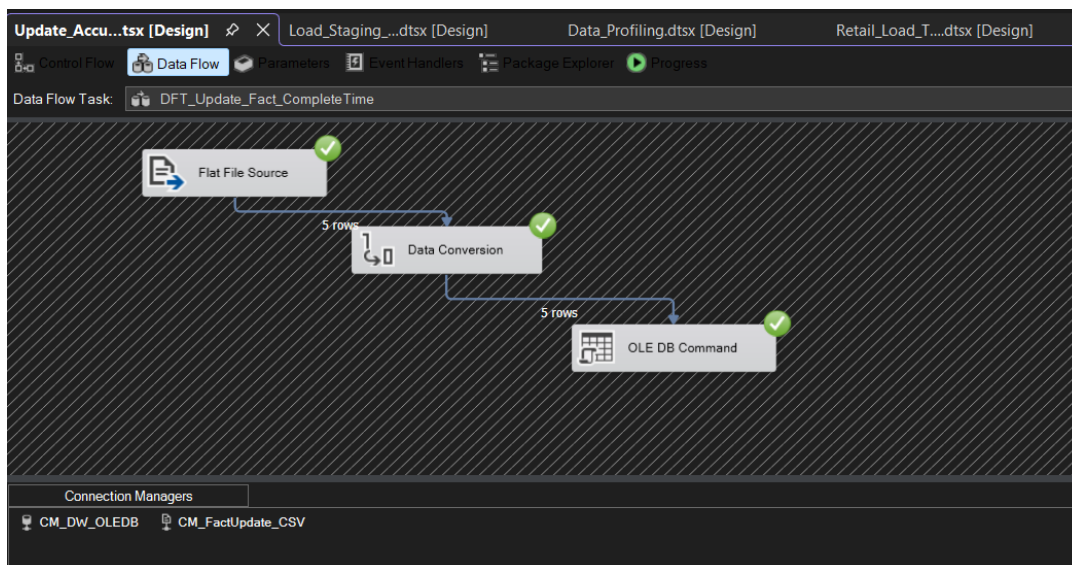
- The Fact_OrderLine table now accurately contains enriched business measures (DiscountAmount, DaysToShip, and ReturnedFlag).
- These calculated fields enable more meaningful analytics such as discount impact analysis, average shipping days by region/category, and return rate calculations.

Now my data warehouse can support advanced reporting on profitability, shipping efficiency, and customer returns, providing valuable insights for operational improvement.

Accumulating fact table control flow



Accumulating fact table Data flow



Retrieve Data from Fact_OrderLine table to check updated data

The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a query window with the following SQL code:

```
1 SELECT TOP 20
2     Order_ID,
3     accm_txn_create_time,
4     accm_txn_complete_time,
5     txn_process_time_hours
6 FROM dbo.Fact_OrderLine
7 WHERE accm_txn_complete_time IS NOT NULL
8 ORDER BY Order_ID;
```

The bottom pane shows the results of the query, which is a table with 5 rows and 4 columns:

	Order_ID	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	CA-2014-115812	2026-03-29 01:50:08.963	2026-04-03 14:30:00.000	133
2	CA-2015-106320	2026-03-29 01:50:08.997	2026-04-06 16:20:00.000	207
3	CA-2016-121755	2026-03-29 01:50:09.000	2026-04-05 11:45:00.000	178
4	CA-2016-152156	2026-03-29 01:50:08.953	2026-04-02 10:00:00.000	105
5	US-2015-108966	2026-03-29 01:50:08.960	2026-04-04 09:15:00.000	152

The status bar at the bottom indicates: "Query executed successfully. DESKTOP-3MQSVBK\MSSQLSERVER... sa (59) Retail_DW | 00:00:00 Row: 1, Col: 1 | 5 rows".